Compositional Design and Operation of Autonomous Multi-Agent Systems

CODAC University Center of Excellence Kick-off

September 24, 2026

Gioele Zardini

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My lab at MIT

muku
411

QUINCY MARKET
THE ALL-NEW NISSAN KICKS

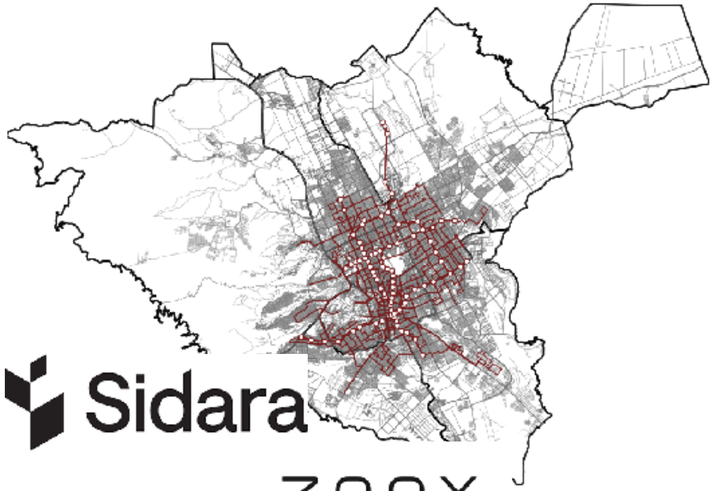
SAMUEL ADAMS

My lab at MIT

Need new tools to model and solve complex systems design optimization problems

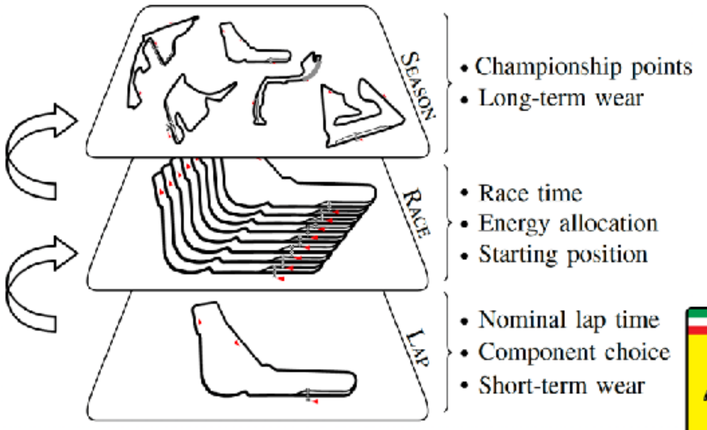
amazonrobotics

SD Guthrie

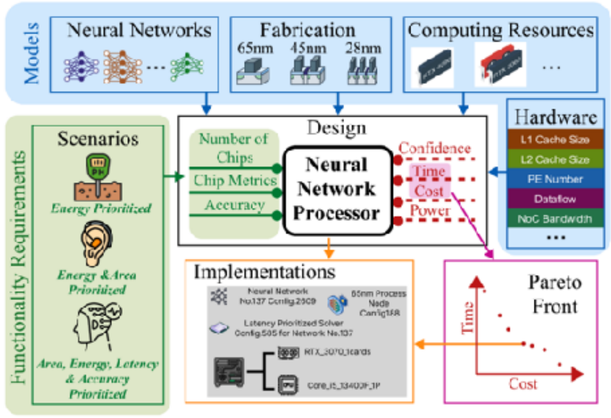


SMI Singapore Maritime Institute
Advance research. Spark innovation.

ABS

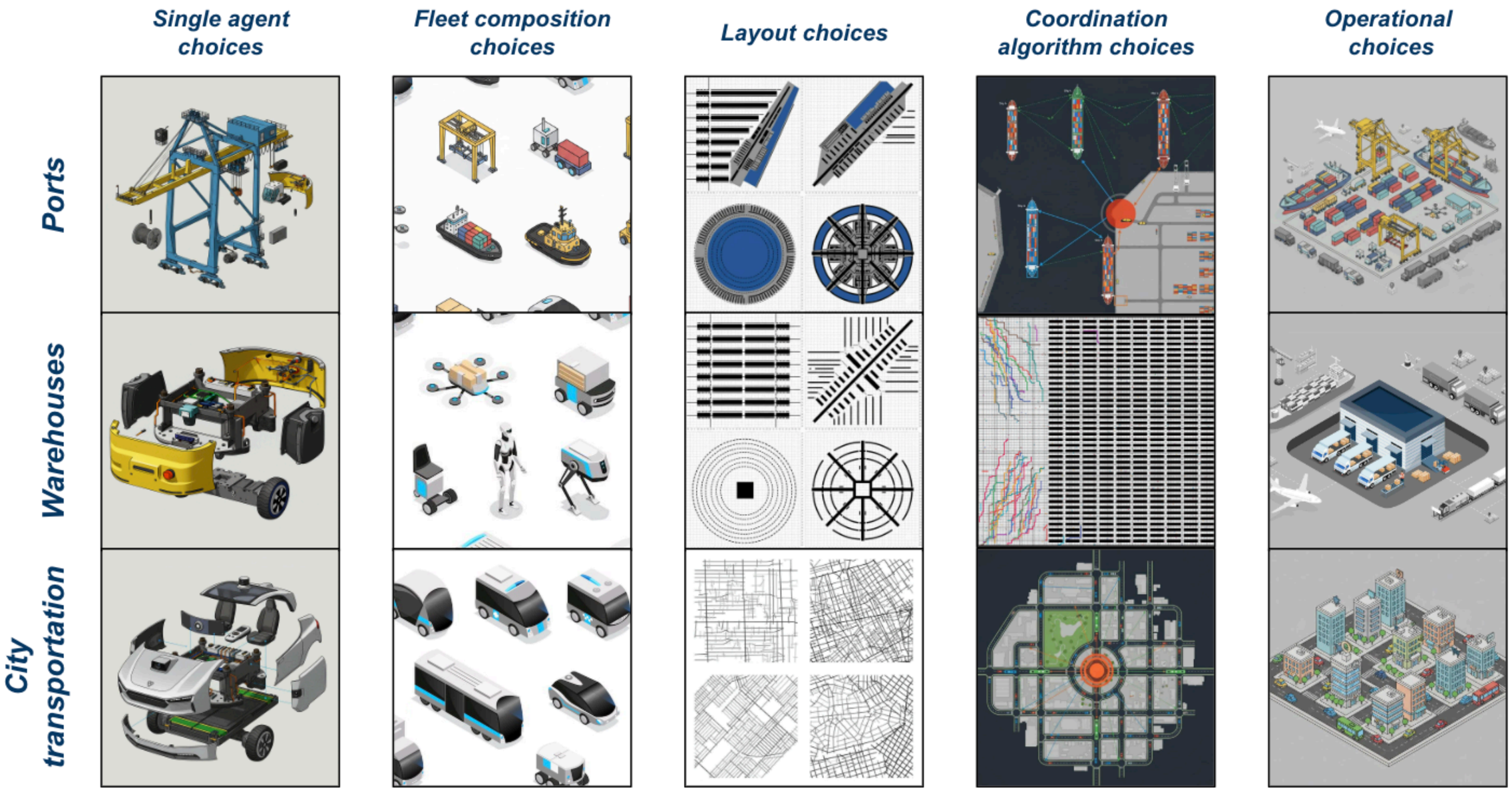


Ferrari



..and more

The success of the technology depends on many components, and their orchestration!



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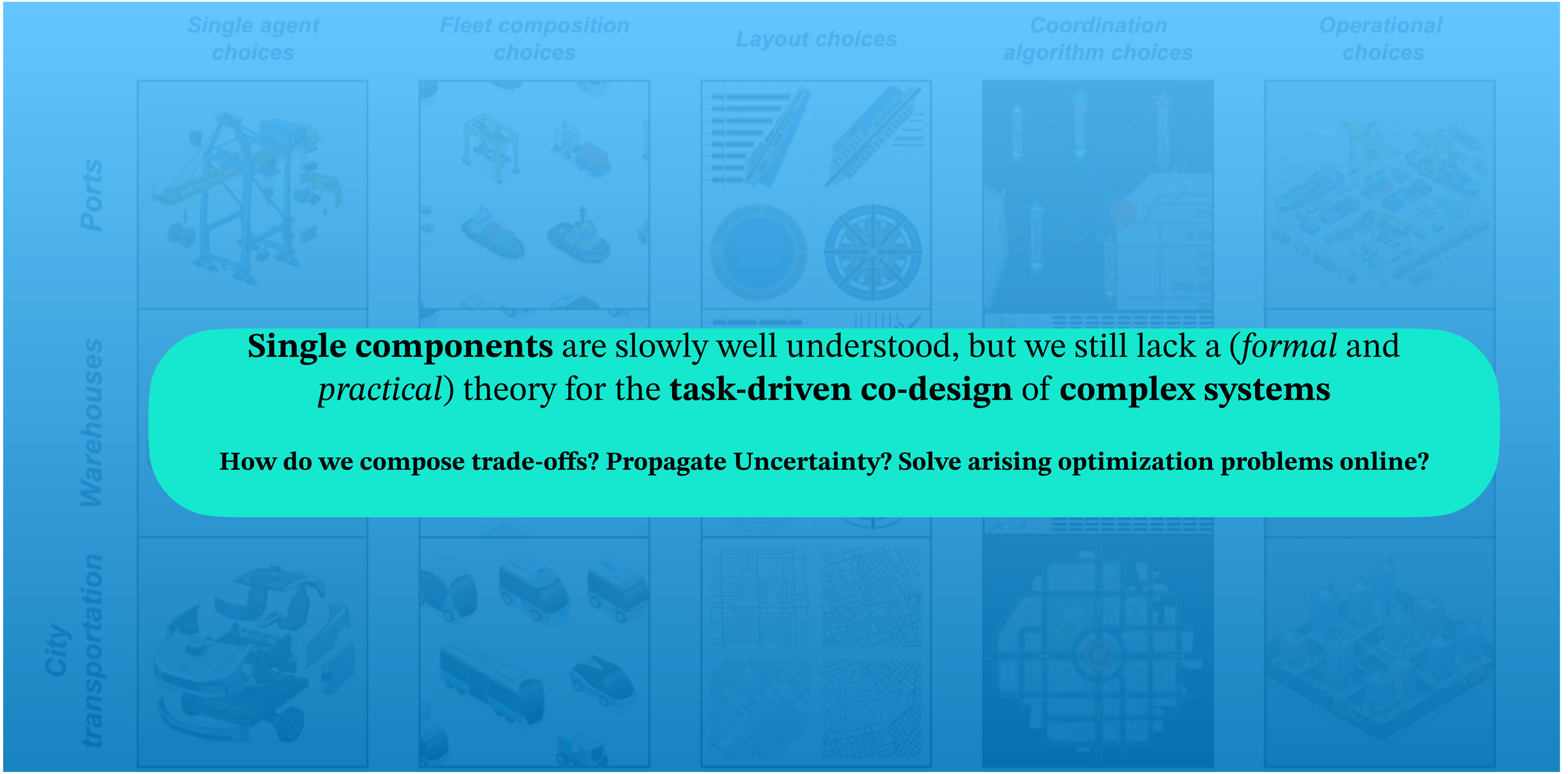


Single components are slowly well understood, but we still lack a (*formal* and *practical*) theory for the **task-driven co-design** of **complex systems**

How do we compose trade-offs? Propagate Uncertainty? Solve arising optimization problems online?

Agenda

- **Motivation**
  - *New challenges of engineering design*
  - *Motivation from autonomy and mobility*
  - *Desiderata for co-design*
- **Monotone Co-Design**
  - *Modeling design problems*
  - *Examples across domains*
  - *Design queries and optimization*
  - *From autonomy to mobility systems*
  - *Latest and greatest*
- **Outlook on ongoing and future research**

Driven by **societal challenges**, we develop **efficient computational tools** to **automate** the **formulation** and **solution** of **large**, **complex system design problems**

Website containing all papers and more pointers:

https://zardini.mit.edu

The vision of automated co-design

minimize
(resources usage)
subject to
(functionality constraints)

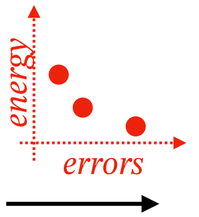
task specification
multi-domain knowledge
design options

“automated designer”

optimal design(s)

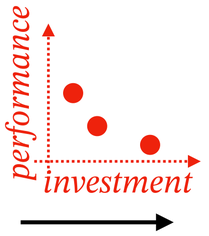
Autonomy co-design

task
robot autonomy, physics
components, algorithms



Mobility co-design

demand
networks, operations, infrastructure
mobility services, policies



Autonomy as the frontier of complexity for the co-design of complex systems

A fleet of autonomous vehicles

=

software **behavior** **coordination**
hardware
actuation **localization** **planning** **invasivity**
sensing
control **interaction** **learning** **liability**
computation
perception **mapping** **regulations**
energetics **communication** **infrastructure**

OMG!
So many components (hardware, software, …),
objectives and **choices** to make!
Nobody understands the **whole** thing!

We forget why we made **choices**, and we are afraid to make **changes** (high failure cost).
We need **faster** design cycles, **nimbler** execution.

anthropomorphization of 21st century engineering malaise

"My dear, it's simple: you lack a *practical* theory of **co-design**!"
Formal,
Quantitative
Intellectually tractable

Your system is just a component in another person's system

Infrastructure level

Infrastructure choices and coordination of services

Service level

Fleet deployment

Platform level

Choice of components and algorithms

Subsystem level

Single component design

Complex systems typically feature multi-stakeholders interactions

Mobility providers (Jamey Tesler, CEO MassDOT)

Liability (Christian Mumenthaler, CEO SwissRe)

Policy makers (Michelle Wu, Mayor Boston)

Academia Tech Developers (Sally Kornbluth, MIT president)

Industry (Laura Major, CTO Motional)

Challenges and requirements for automated co-design of complex systems

Complexity when designing complex systems

Large systems
- Many components, scales
- Heterogeneous natures
- Multiple objectives

Strategic interactions
- Many agents
- Heterogeneous interactions
- Conflicts/collaborations

- **Formal, domain-independent**
- **Computationally tractable**
  - Need to compute solutions efficiently
- **Compositional, hierarchical**
  - My system is a component of somebody else's system
- **Collaborative**
  - Pooling knowledge from experts across fields.
- **Intellectually tractable**
  - Not exclusively accessible to system architects
- **Continuous/adaptive**
  - Design is not static: it should be reactive to changes in goals and contexts

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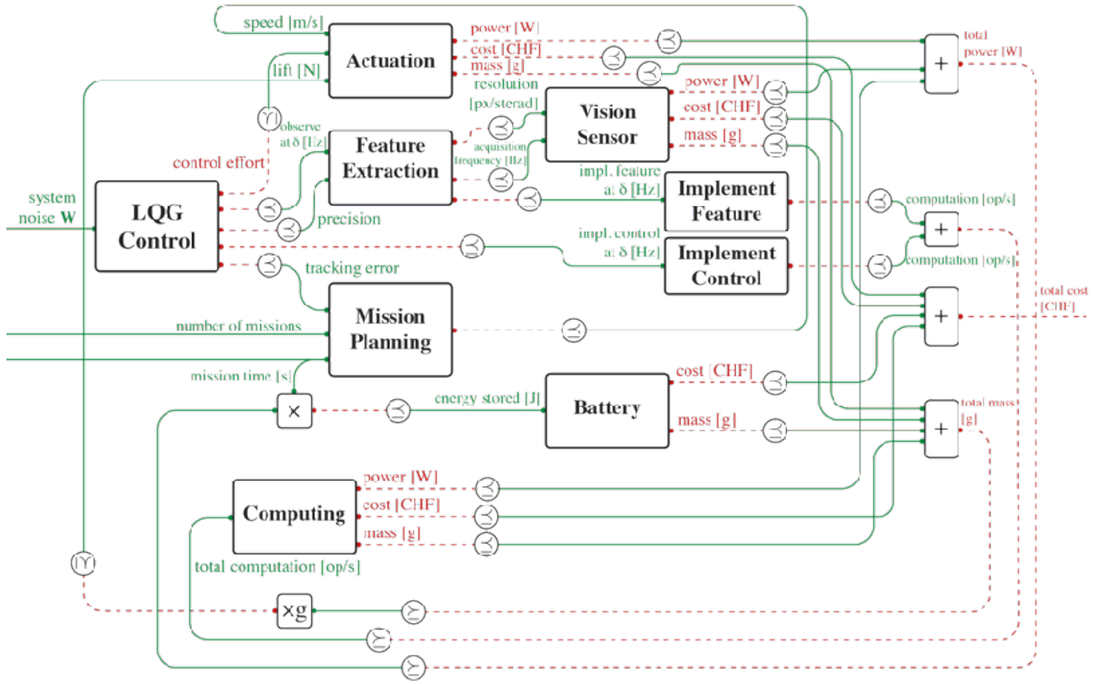
A new approach to multi-disciplinary engineering “co”-design

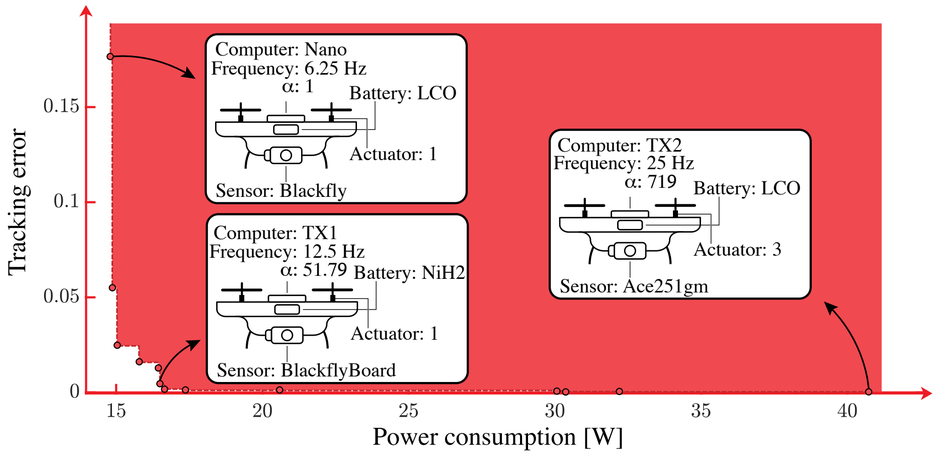
- A new approach to **collaborative**, **computational**, **compositional**, **continuous** design designed to work **across fields** and **across scales**.
- Leverages **domain theory**, **applied category theory**, and **optimization**
- Roadmap:
  - Defining **“design problems”** for **components.**
  - Modeling **co-design constraints** in a complex **system**.
  - **Efficient** solution to design queries.

“Co-design diagram”



optimization for a task

Pareto front of optimal designs



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a “pro” box

To be published by Cambridge University Press
https://bit.ly/3qQNrdR

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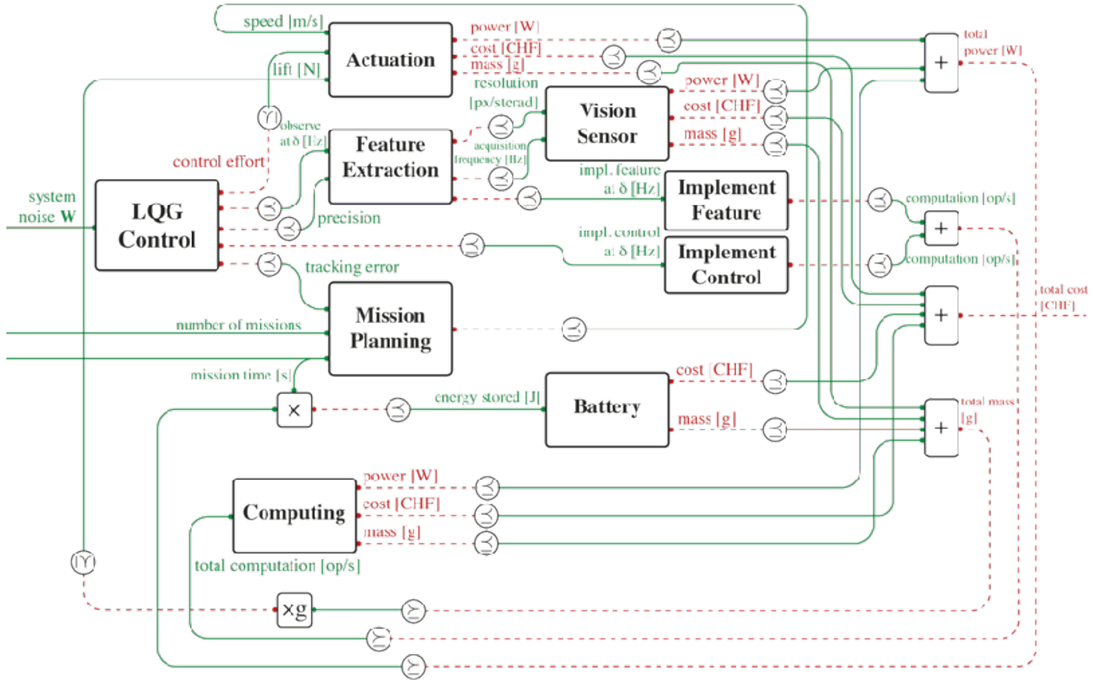

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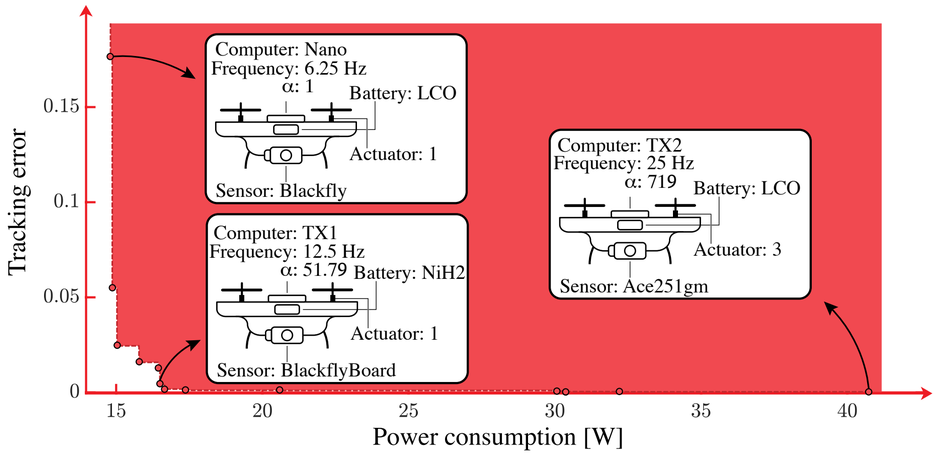
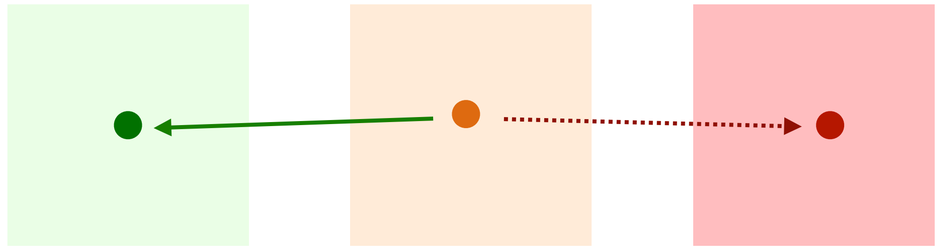


An abstract view of design problems

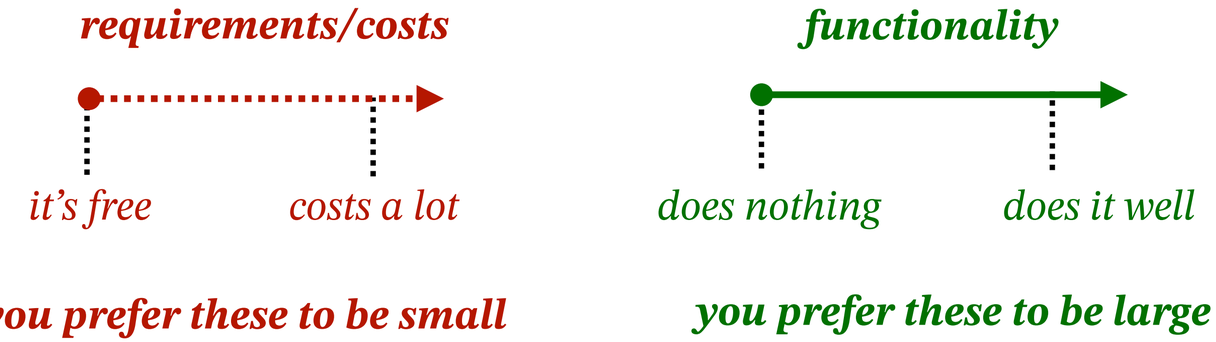
- Across fields, design or synthesis problems are defined with **three spaces**:
  - **implementation space**: the **options** we can choose from;
  - **functionality space**: what we need to **provide**/**achieve**;
  - **requirements/costs space**: the **resources** we need to have available;



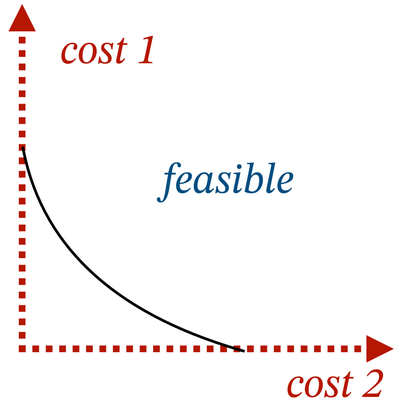
functionality *(provided)*	***implementations***	***costs, resources*** *(required)*
desired behavior	*choices*	*budget*
specifications	*plans*	*requirements*
objectives	*blueprints*	*dependencies*
guarantees	*decision variables*	*assumptions*
conclusions		
"function"	*"form"*	*"function"*

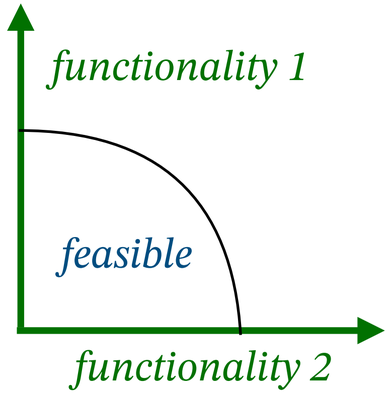
Trade-offs are everywhere: embrace them!

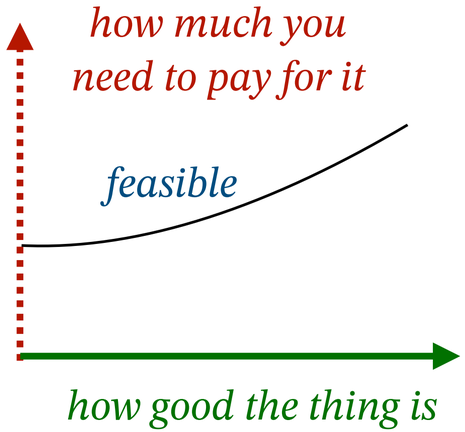
- We distinguish (semantically) between **functionality** and **requirements/costs.**



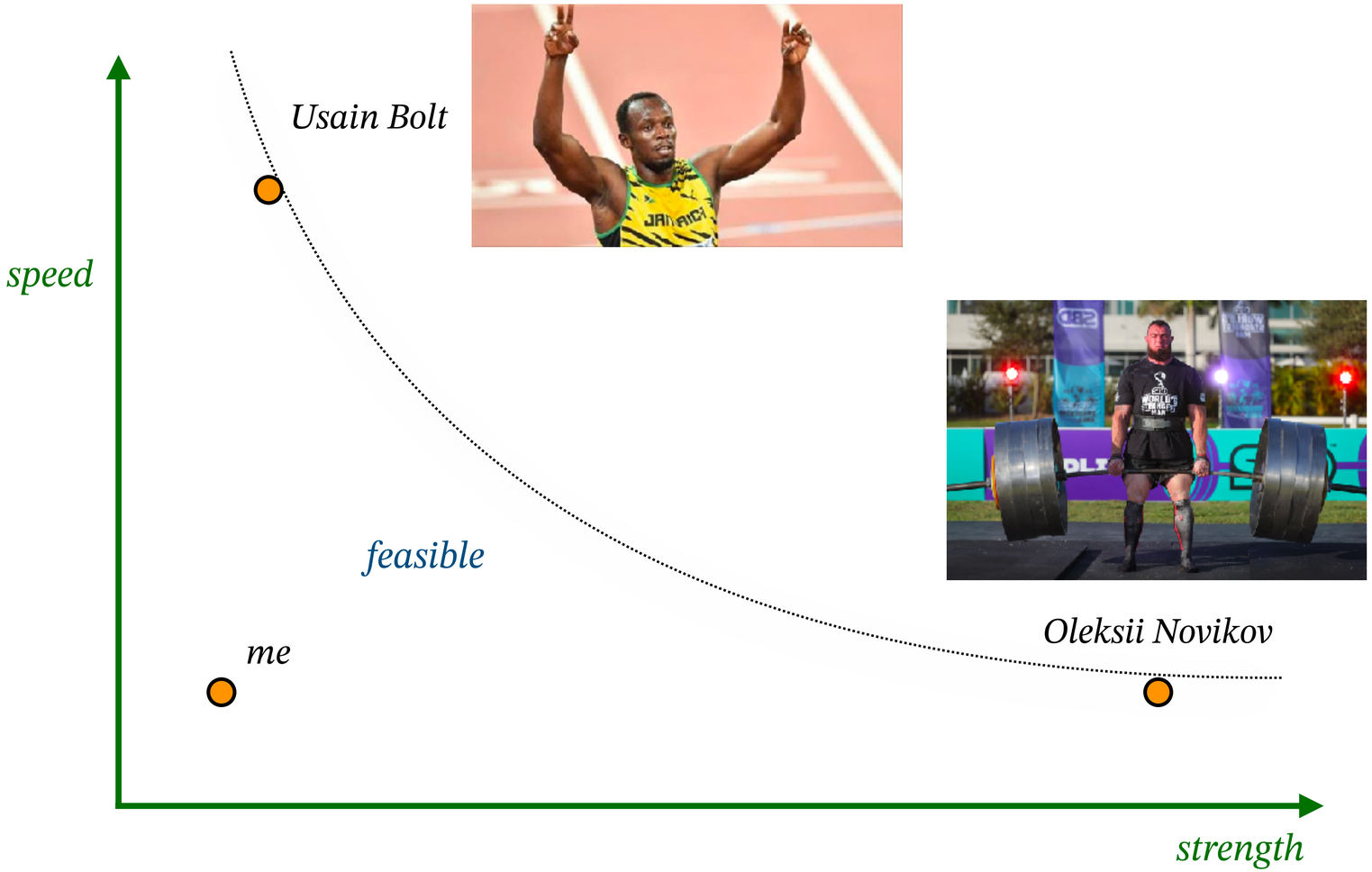
- Open an engineering book. Find the graphs talking about "achievable performance". What colors should the axes be? Classify into one of these:





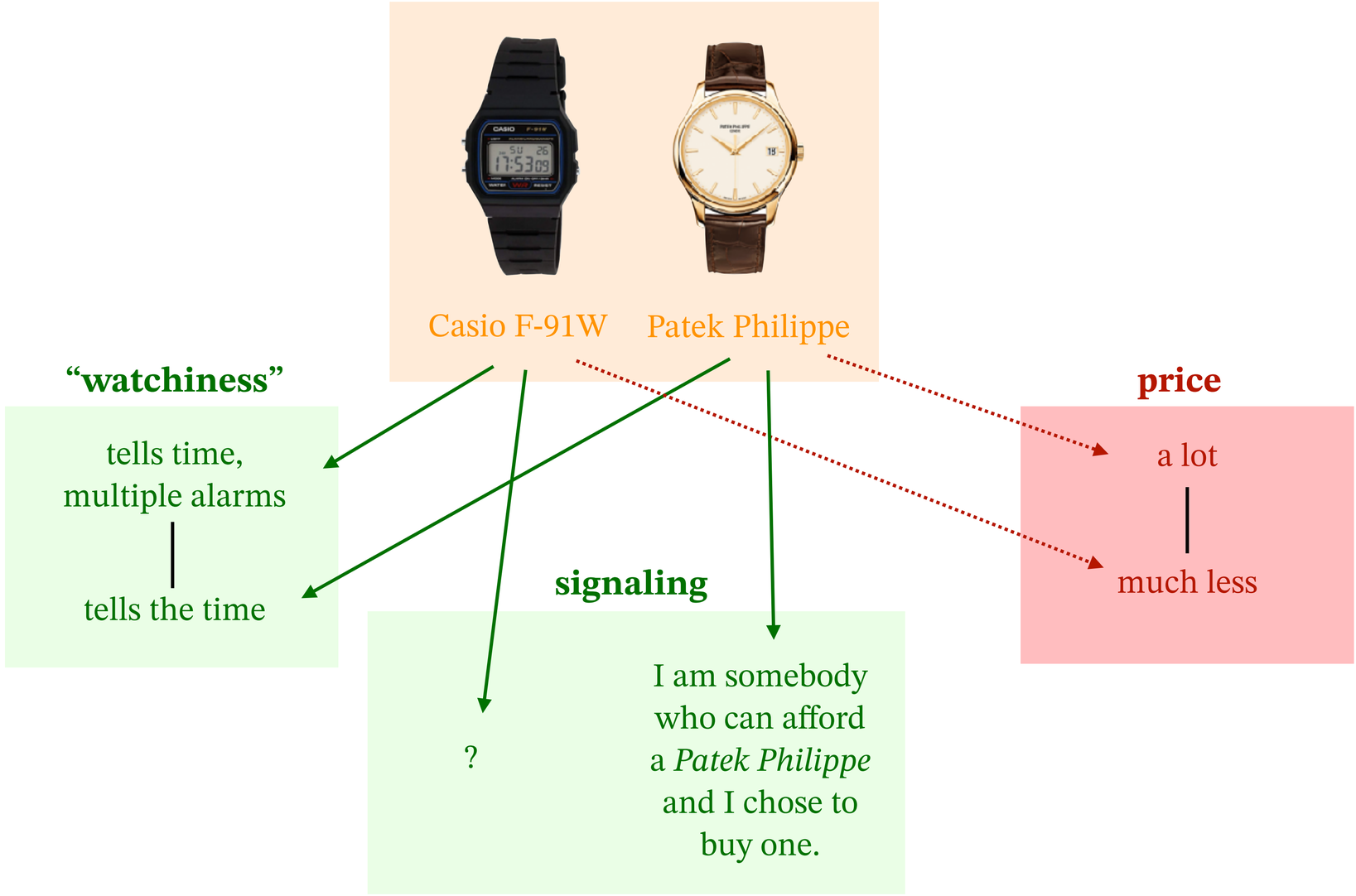


Trade-offs for the human body



Function beyond the strictly technical meaning

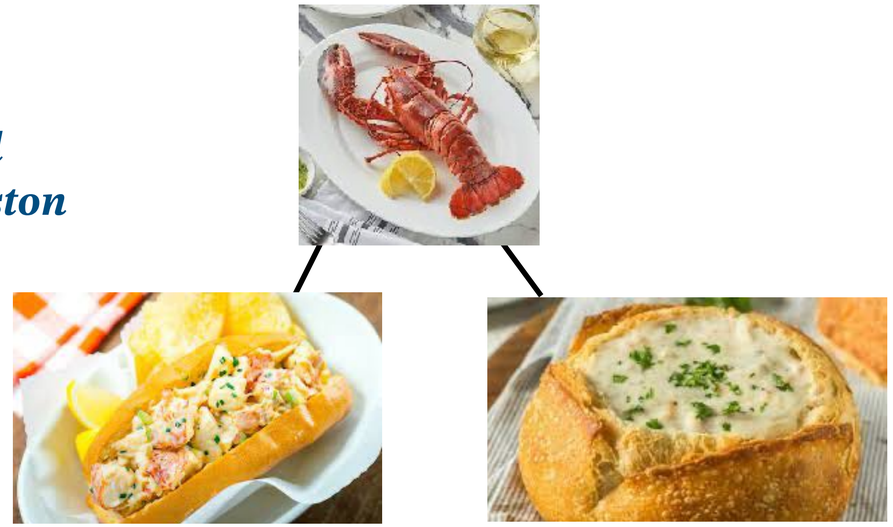
- **Law of successful products**: at equilibrium, in an efficient and free market, no product completely dominates another by both functionality and costs.
  (Otherwise, the dominated product would not sell.)



Partially ordered sets model trade-offs, across fields

- Posets model standard costs in engineering $\langle \mathbb{R}_{\geq 0}, \leq \rangle$, $\langle \mathbb{N}, \leq \rangle$
- … but also enable **richer** cost structures, with **incomparable** elements

A poset of food preferences in Boston

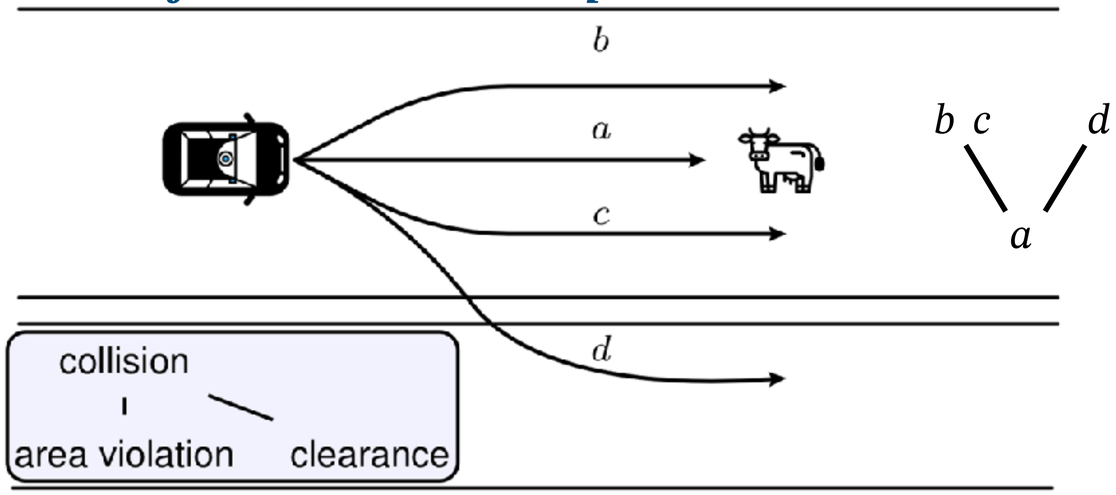


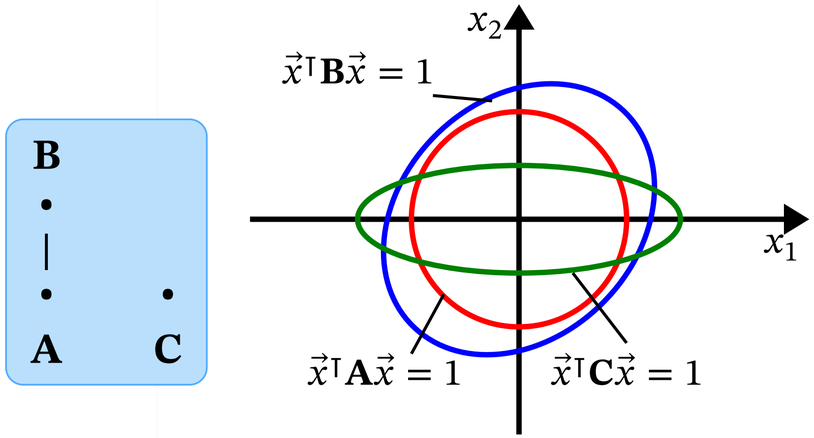
A poset of positive-definite matrices

$$\frac{\mathbf{A} \preceq_{\mathsf{PDM}(n)} \mathbf{B}}{\vec{x}^\intercal \mathbf{A} \vec{x} \leq \vec{x}^\intercal \mathbf{B} \vec{x} \quad \forall \vec{x} \in \mathbb{R}^n}$$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3/4 & -1/8 \\ -1/8 & 3/4 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$$

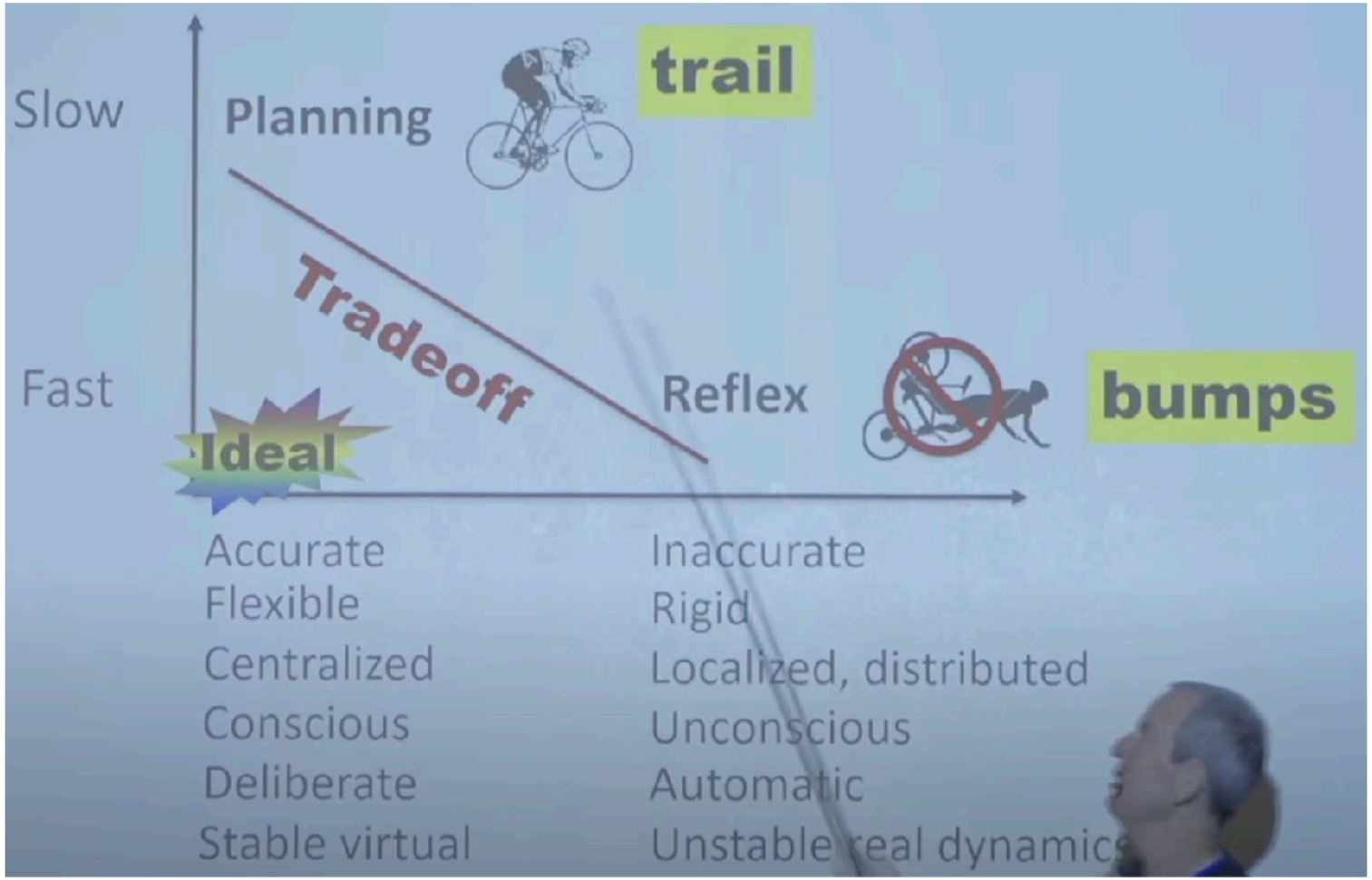
Posets of rules, which induce priorities over behaviors





[RAL'22, CDC'22, IROS'21, ECC'21]

Trade-offs



From John Doyle's talks

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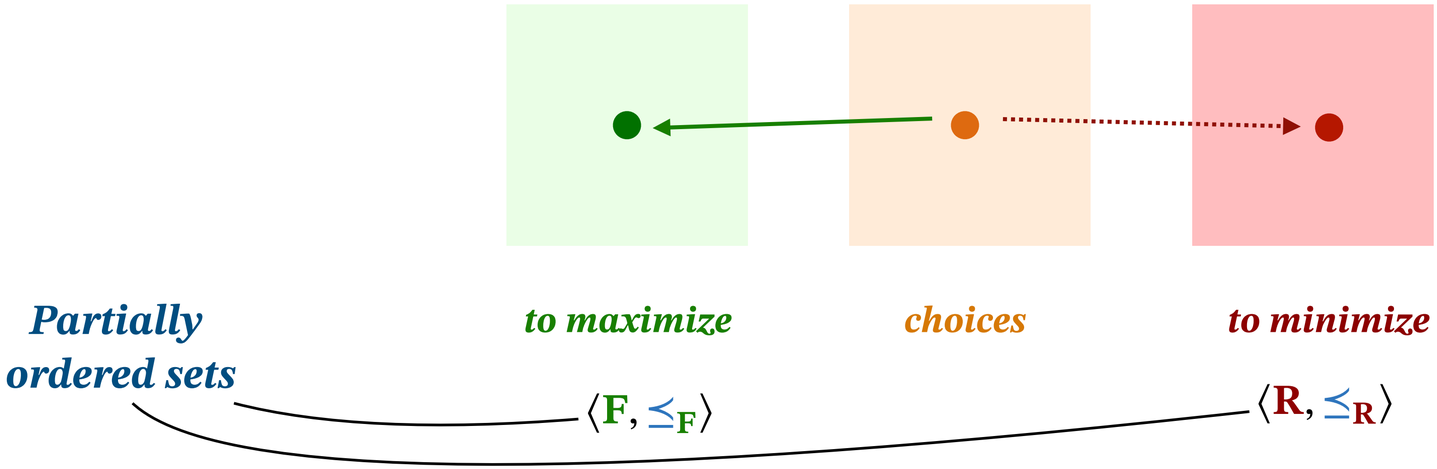
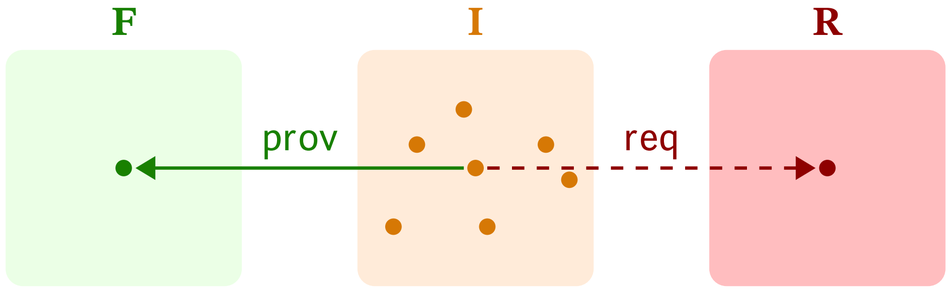
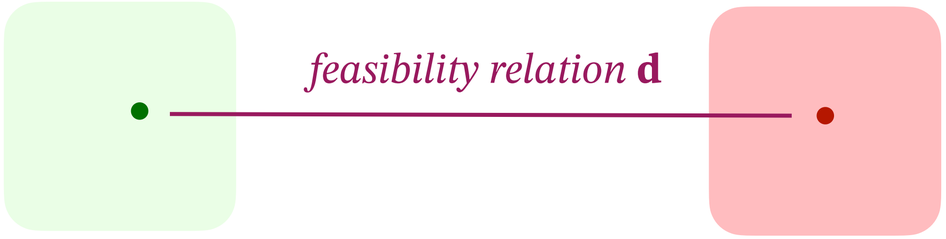


Transparent vs black-box models

- The "**Design Problems with Implementations" model** is a "**transparent" model:**



- **DP** model: **direct feasibility relation** between functionality and resources ("black box") as a monotone map:



$$\mathbf{d} : \mathbf{F}^{\mathrm{op}} \times \mathbf{R} \to_{\mathbf{Pos}} \mathbf{Bool}$$
$$\langle f^*, r \rangle \mapsto \exists i \in \mathbf{I} : (f \leq_{\mathbf{F}} \mathsf{prov}(i)) \wedge (\mathsf{req}(i) \leq_{\mathbf{R}} r)$$

... a "boolean profunctor"

- **Monotonicity**:
  - **Lower functionality** does **not** require **more resources**;
  - More **resources** do not provide **less functionality.**

Co-design enables a rich class of model population techniques

- “**Catalogues**”: off-the-shelf designs.

	Spark	Phantom 3 Std	Phantom 4 Adv	Phantom 4 Pro	Mavic	Inspire
Flight time	16 mins	25 mins	30 mins	30 mins	27 mins	27 mins
Top Speed	31 mph (50 km/h)	36 mph (58 km/h)	45 mph (72 km/h)	45 mph (72 km/h)	40 mph (65 km/h)	58 mph (94 km/h)
Range	1.2 miles (2 km)	0.6 miles (1 km)	4.3 miles (7 km)	4.3 miles (7 km)	4.3 miles (7 km)	4.3 miles (7 km)
Camera	12-MP stills 1080p video	12-MP stills 2704 x 1520p video	20-MP stills 4K 60fps video	20-MP stills 4K 60fps video	12-MP stills 4K video	20.8-MP stills 4K/5K video
Size	5.6 x 5.6 x 2.1 in (14.3 x 14.3 x 5.5 cm)	13.8 in diagonal (350 mm)	13.8 in diagonal (350 mm)	13.8 in diagonal (350 mm)	13.2 in diagonal (350 mm)	16.8 x 12.5 x 16.7 in (42.7 x 31.7 x 42.5 cm)
Takeoff weight	11.6 oz (330 g)	2.6 lb (1.2 kg)	3 lb (1.4 kg)	3 lb (1.4 kg)	1.6 lb (743 kg)	8.8 lb (4 kg)
Other features	Follow me, Return home, Obstacle avoidance, FPV	Follow me, Return home	Follow me, Return home, Obstacle avoidance	Follow me, Return home, 3 Direction Obstacle avoidance	Follow me, Return home, Obstacle avoidance, folding arms	Obstacle avoidance, Spotlight Pro/Broadcast/ Composition mode
Price	US$499	US$499	US$1,349	US$1,499	US$999	US$2,999 ($6,198 with camera/gimbal)

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- “**First-principles**”: analytical relations.

mission energy ≥ *mission duration* x *power consumption*

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- “**Data-driven**”, “**on-demand**”
  - The optimization will only ask for a **sequence** of data points. The model is constructed **incrementally**.
  - Opens the door to **experiments**, black-box **simulations**, solutions of **optimization problems**.

Design problems arise naturally in many domains, across scales

demand

investments

externalities

obs. noise

syst. noise

LQG

tracking error

effort

speed [rad/s]

torque [Nm]

cost [$]

mass [kg]

current [A]

guarantees

assumptions

speed [m/s]

task

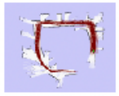
fix cost

operational cost

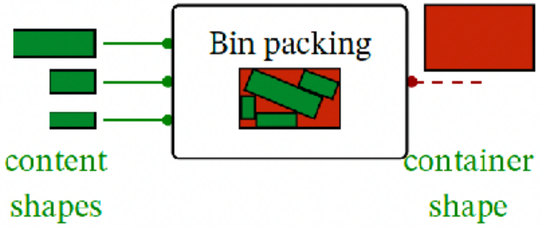
accuracy

robustness

GMapping



nr. particles



area to cover [m2]

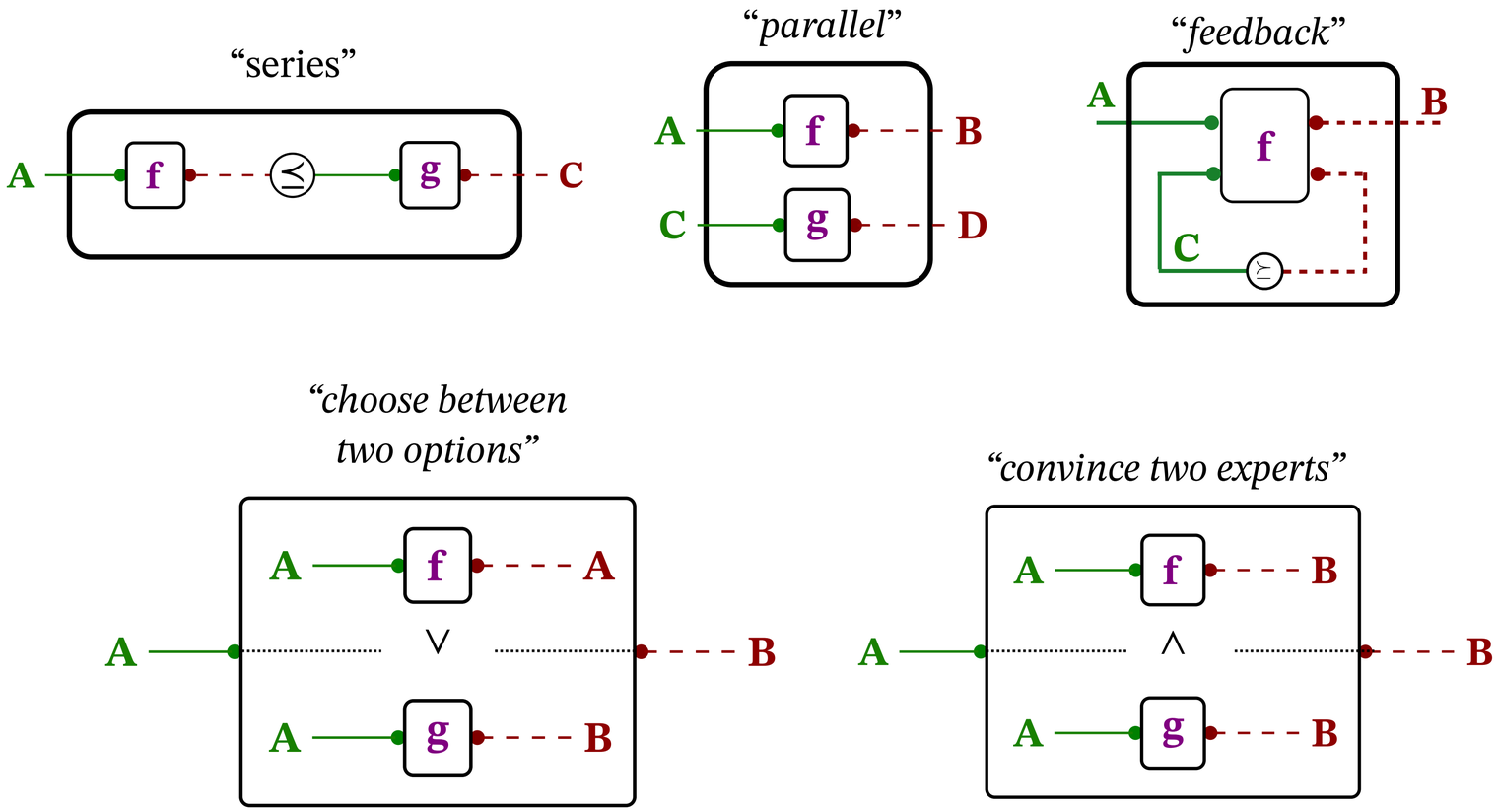
agents' speed [m/s]

agents

time [s]

total cost [$]

Design problems can be composed in various ways, preserving properties



- The **composition** of any two **DPs** returns a **DP** (closure)
- Very practical tool to **decompose** large **problems** into **subproblems**

> **"A system is composed of components; a component is something you understand."**
>
> — Howard Aiken (1900-1973)

*There is a category **DP** which is traced monoidal, and locally posetal*

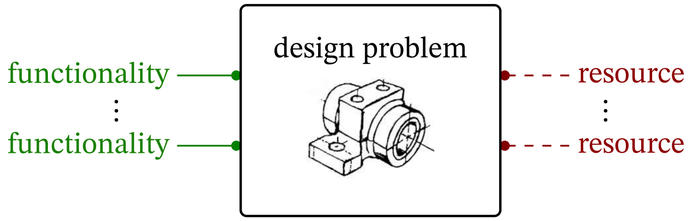
✓ ***Formal***
Compositional/hierarchical

Multiple queries from the same design problem

- Two basic design queries are:

Given the functionality to be provided,
what are the **minimal resources** required?

FixFunMinReq



FixReqMaxFun

Given the resources that are available, what is
the **maximal functionality** that can be provided?

Multiple queries from the same design problem

- Two basic design queries are:

Given the functionality to be provided,
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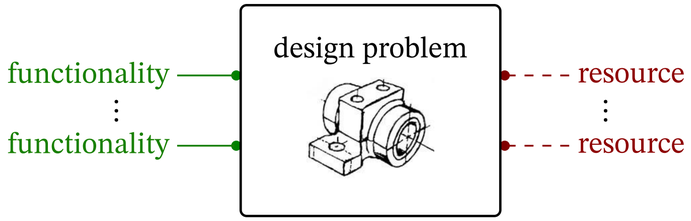
FixFunMinReq



FixReqMaxFun

Given the resources that are available, what is
the **maximal functionality** that can be provided?

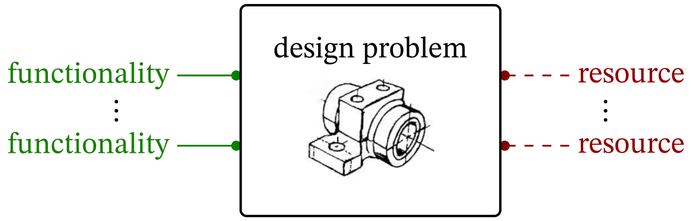
- The two problems are **dual**

Multiple queries from the same design problem

- Two basic design queries are:

Given the functionality to be provided,
what are the **minimal resources** required?

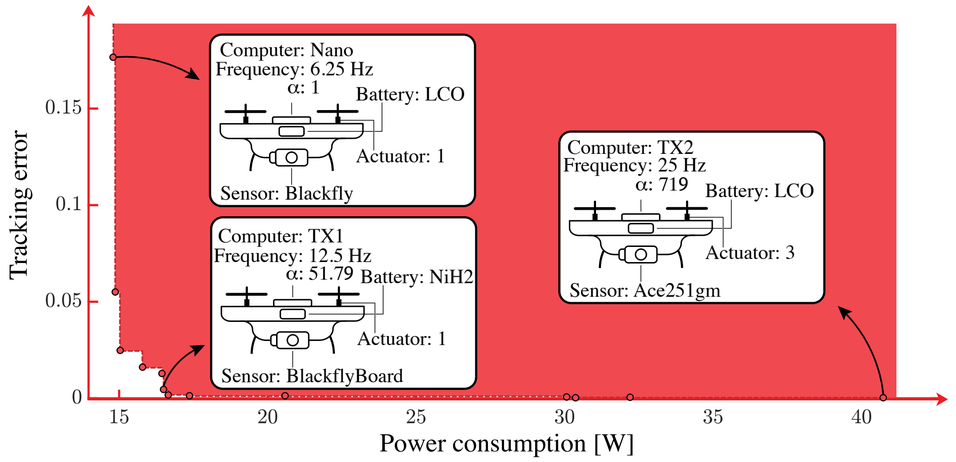
FixFunMinReq



FixReqMaxFun

Given the resources that are available, what is
the **maximal functionality** that can be provided?

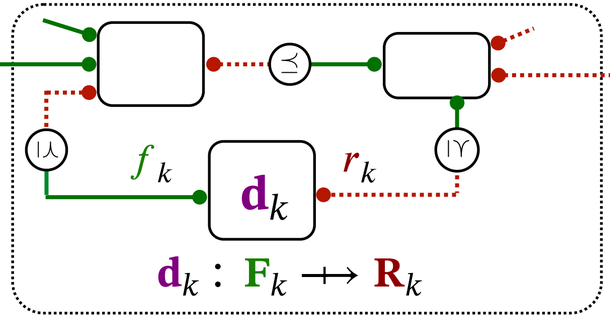
- We are looking for:
  - A map from functionality to **upper sets** of feasible resources: $h : \mathbf{F} \to \mathcal{U}\mathbf{R}$
  - A map from functionality to **antichains** of minimal resources: $h : \mathbf{F} \to \mathcal{A}\mathbf{R}$



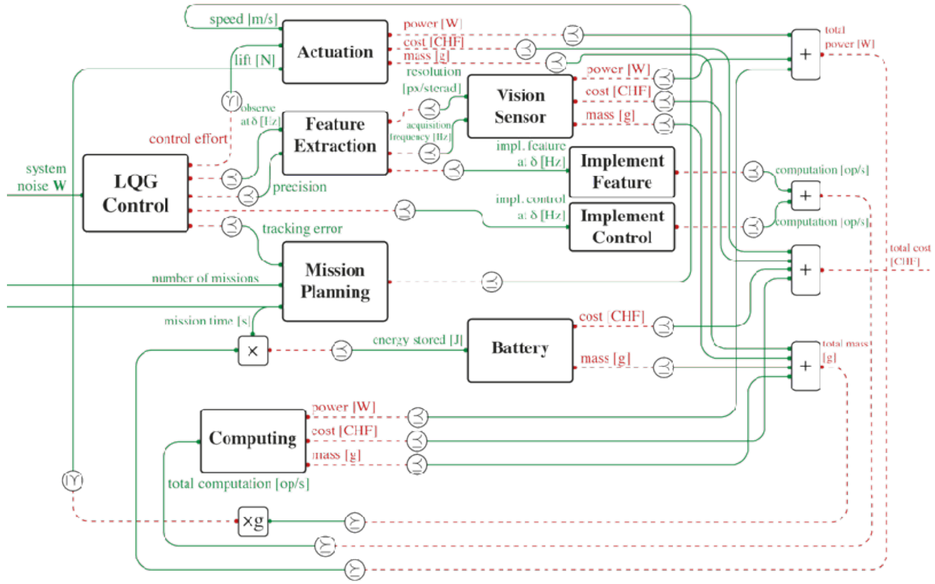
A new family of compositional optimization problems

- This is the semantics of **FixFunMinReq** as a **family of optimization problems.**

chosen by user $\overline{f}$

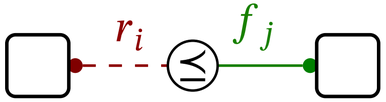


$\overline{r}$ **to minimize**



variables $f_k \in \langle \mathbf{F}_k, \preceq_{\mathbf{F}_k} \rangle$ $r_k \in \langle \mathbf{R}_k, \preceq_{\mathbf{R}_k} \rangle$

constraints *for **each node**:* *for **each edge:***



$\mathbf{d}_k(f_k^*, r_k) = \top$ $r_i \preceq f_j$

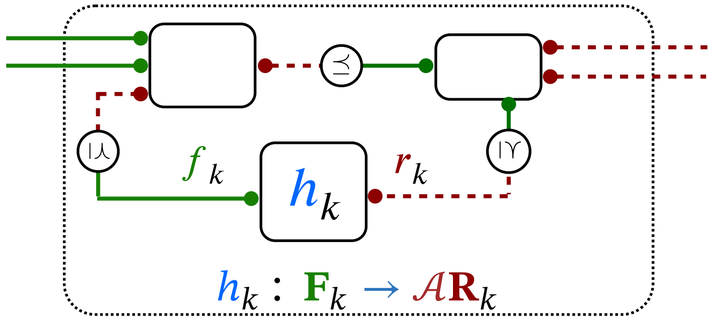
component feasibility *co-design constraint*

! **not convex**
! **not differentiable**
! **not continuous**
! **not even defined on continuous spaces**

objective $\underset{\preceq}{\text{Min}}\, \bar{r}$

Compositional solution of design problem queries

- Suppose that we are given the map $h_k : \mathbf{F}_k \to \mathcal{A}\mathbf{R}_k$ for all nodes in the co-design graph



- Can we find the map $h : \mathbf{F} \to \mathcal{A}\mathbf{R}$ for the entire graph?

✔ ***Computationally tractable****

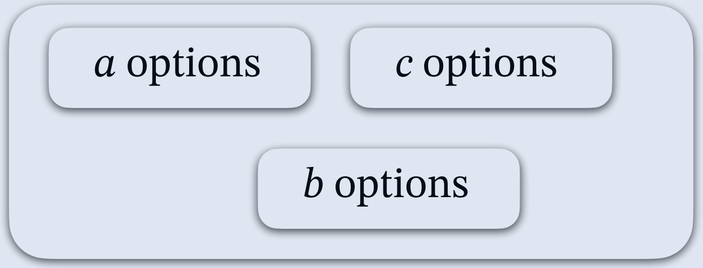
- **Compositional approach**: just need to work out the composition formulas for all operations

solution(**composition**(a, b)) = **composition**(**solution**(a), **solution**(b))

... a functor between a category of problems and one of solutions

- The set of **minimal** feasible resources can be obtained as the **least fixed point** of a monotone function in the space of anti-chain

- We have a **complete solution**: guaranteed to find the set of **all** optimal solutions (if empty, **certificate of infeasibility**)

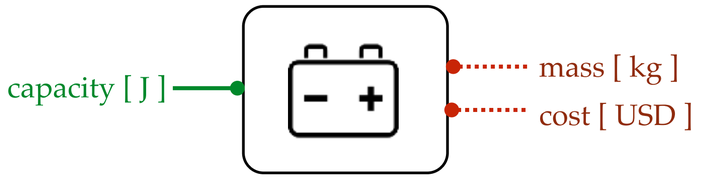


- The complexity is **not combinatorial in the number of options** for each component
- The complexity depends on the **complexity of the interactions**: the co-design **constraints**

$O(a + b + c)$

User-friendly interfaces

- "**Catalogues**": already available designs

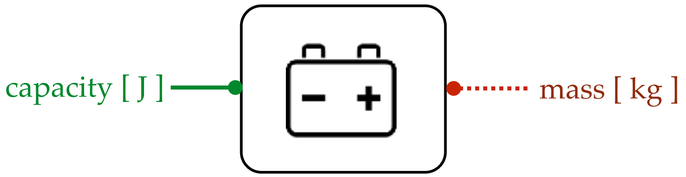



```
catalogue {
    provides capacity [J]
    requires mass [g]
    requires cost [USD]

    500 kWh ⟷ model1 ⟼ 100 g, 10 USD
    600 kWh ⟷ model2 ⟼ 200 g, 200 USD
    600 kWh ⟷ model3 ⟼ 250 g, 150 USD
    700 kWh ⟷ model4 ⟼ 400 g, 400 USD
}
```

... and a solver

- "**First-principles**": analytical relations.




```
mcdp {
    provides capacity [J]
    requires mass [kg]

    specific_energy_Li_Ion = 500 Wh / kg

    required mass >= provided capacity / specific_energy_Li_Ion
}
```

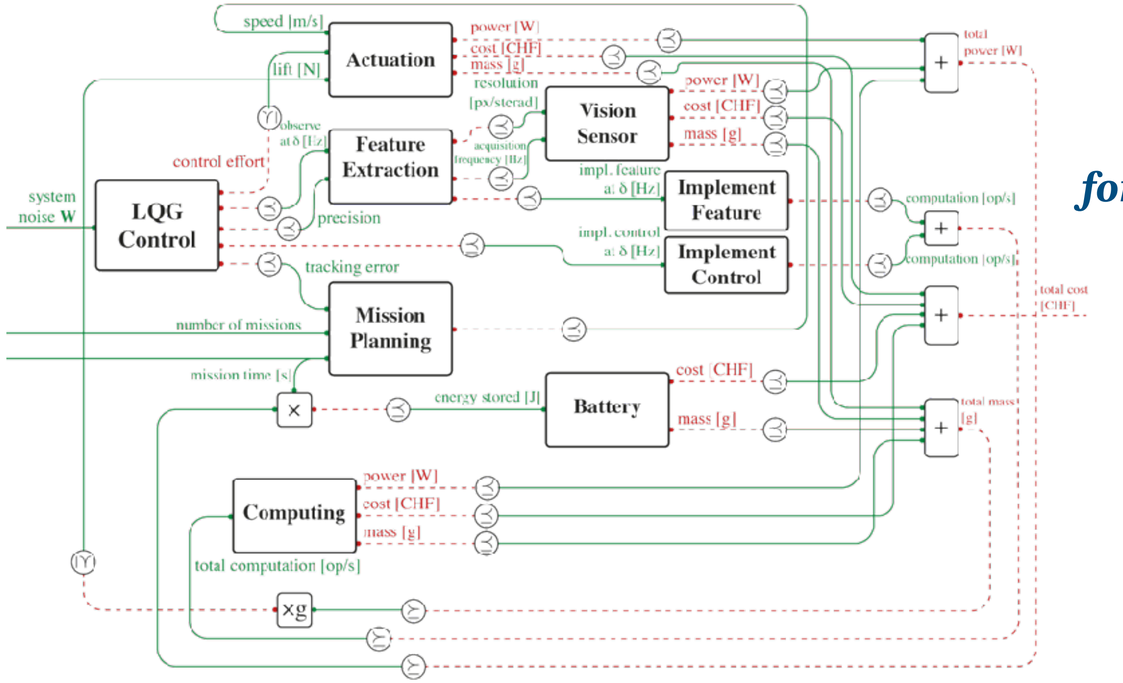
```
mcdp {
    provides lift [N]
    requires power [W]
    c = 10.0 W/N²
    required power ≽ c · provided lift²
}
```

A systematic process for task-driven co-design of complex systems

- A new approach to **co-design** designed to work **across fields** and **across scales**.
- What we have seen:
  - Defining **"design problems"** for **components.**
  - Modeling **co-design constraints** in a complex **system**.
  - **Efficient** solution to design queries.

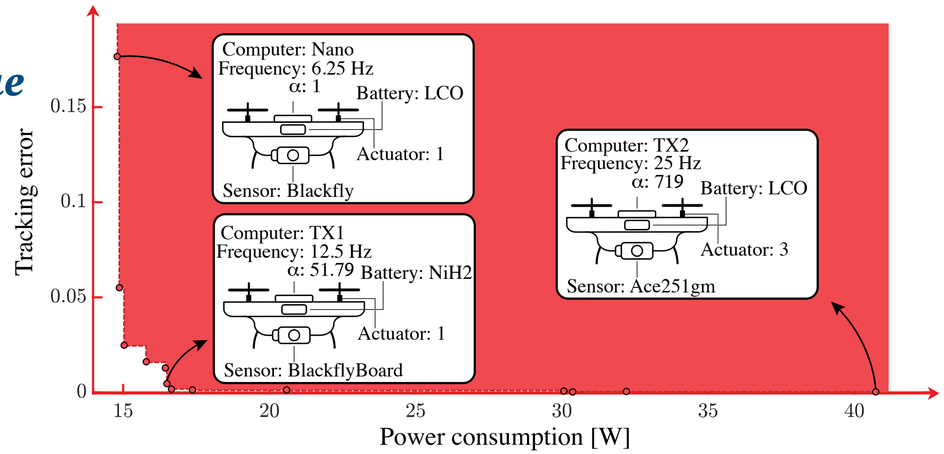
- **Modeling** approach
- Actual **implementation**:
  - Coming up with the skeleton/diagram
  - Populating the models

"Co-design diagram for a drone"

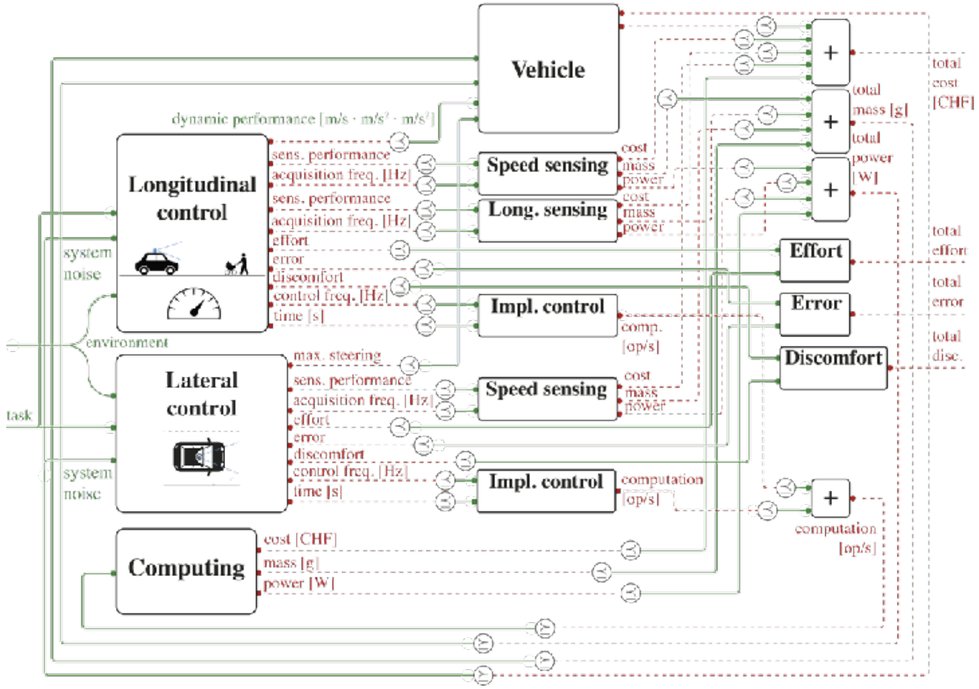


optimization for a search-and-rescue task

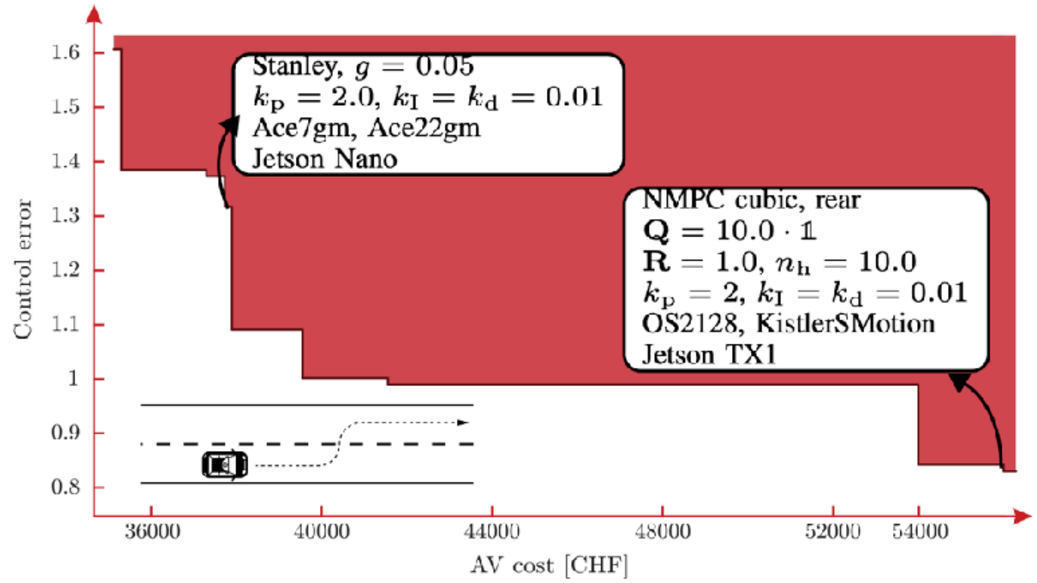
Pareto front of optimal designs

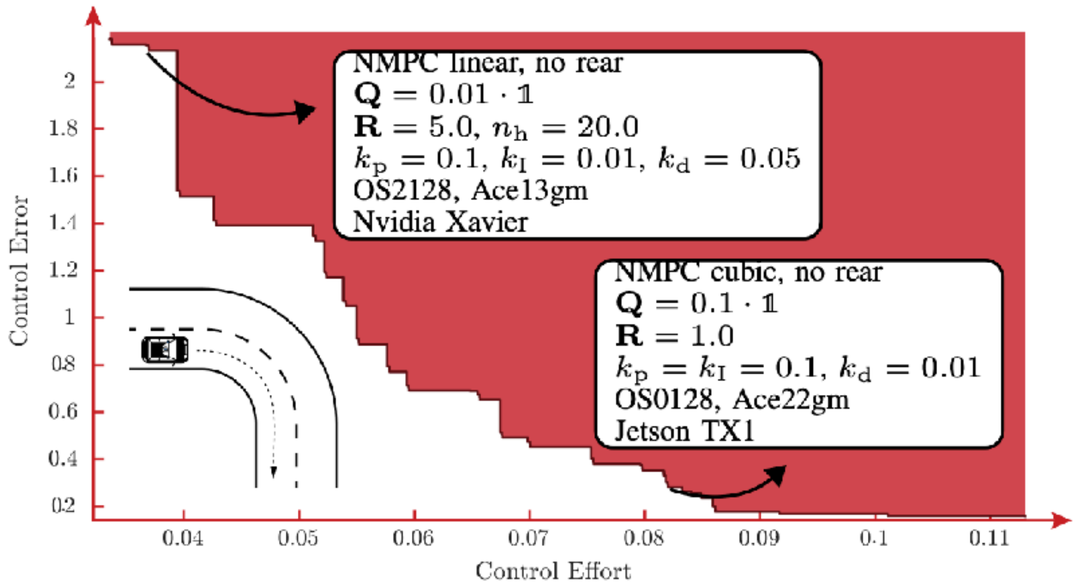


We can find optimal designs, with insights at heterogeneous abstraction levels

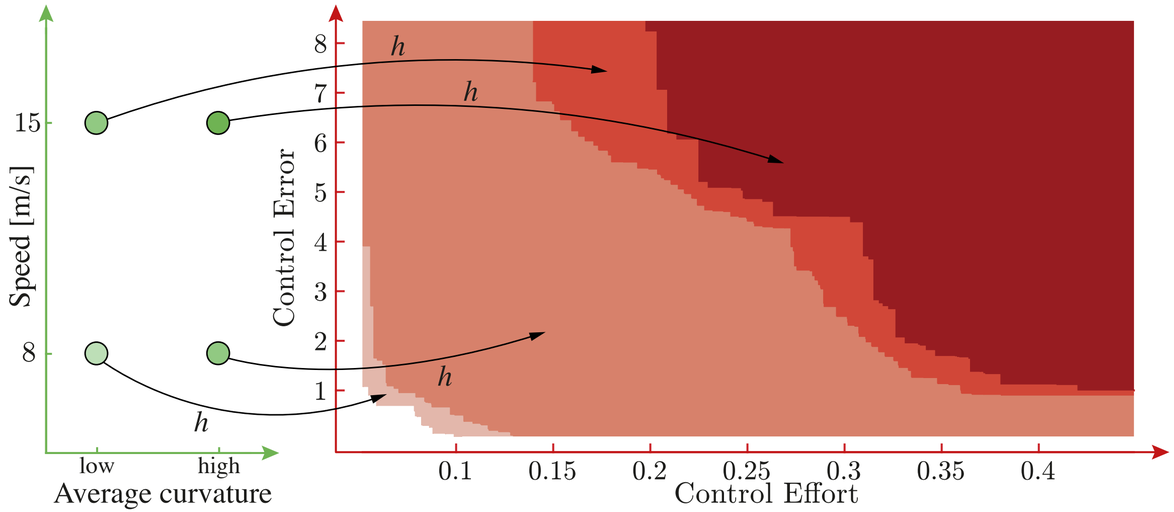


Fix an environment
Fix a task



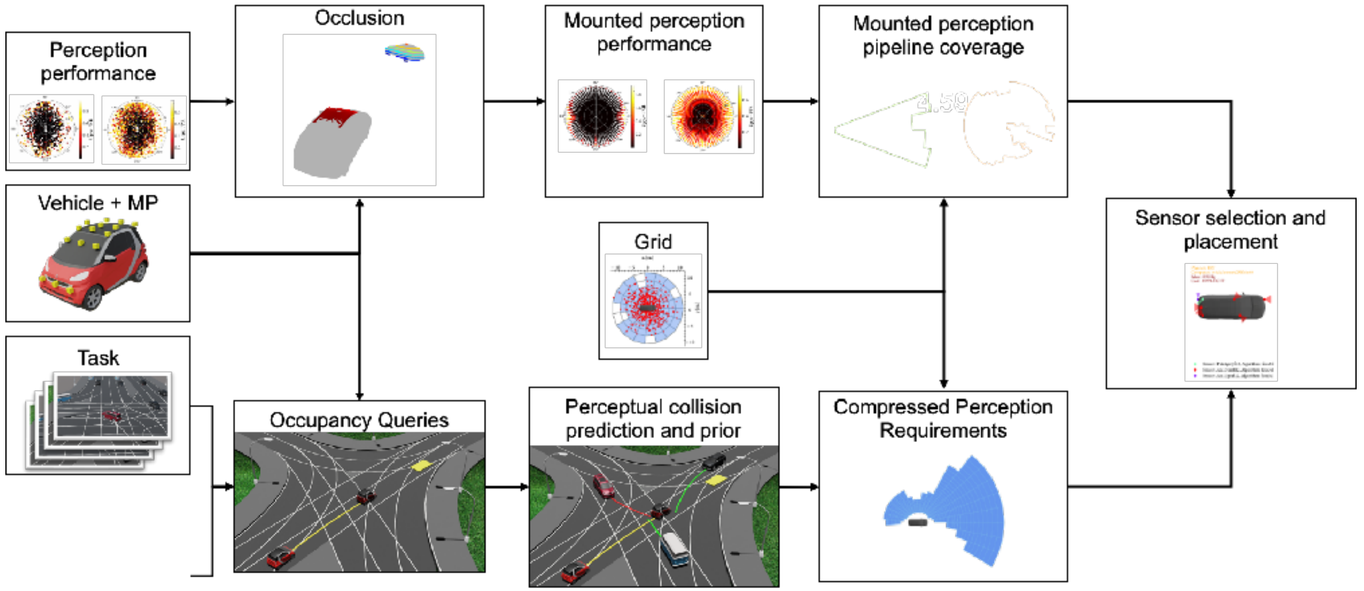


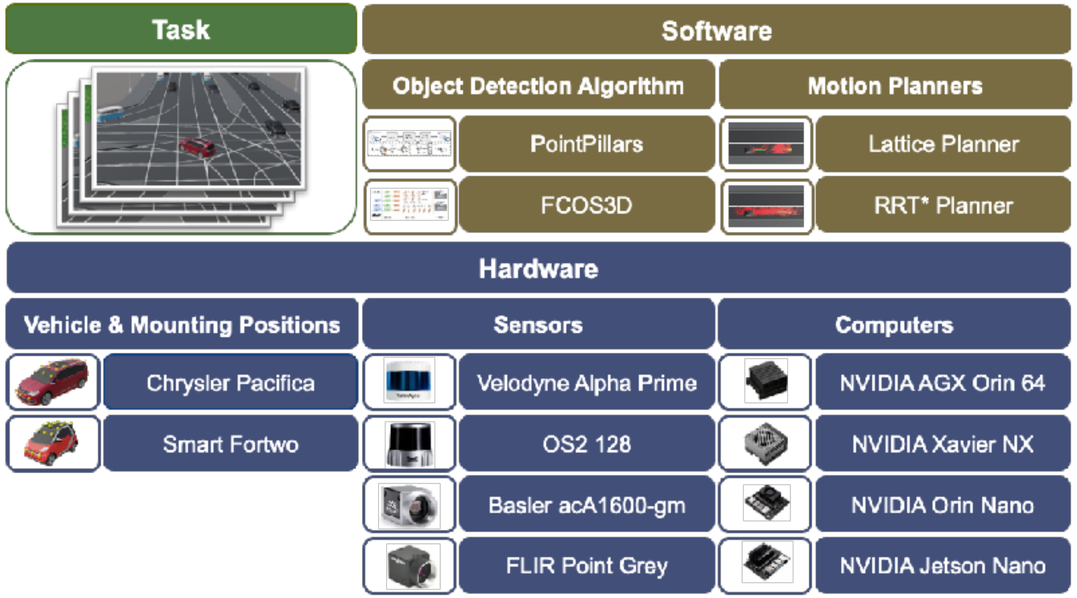
Monotonicity in task complexity

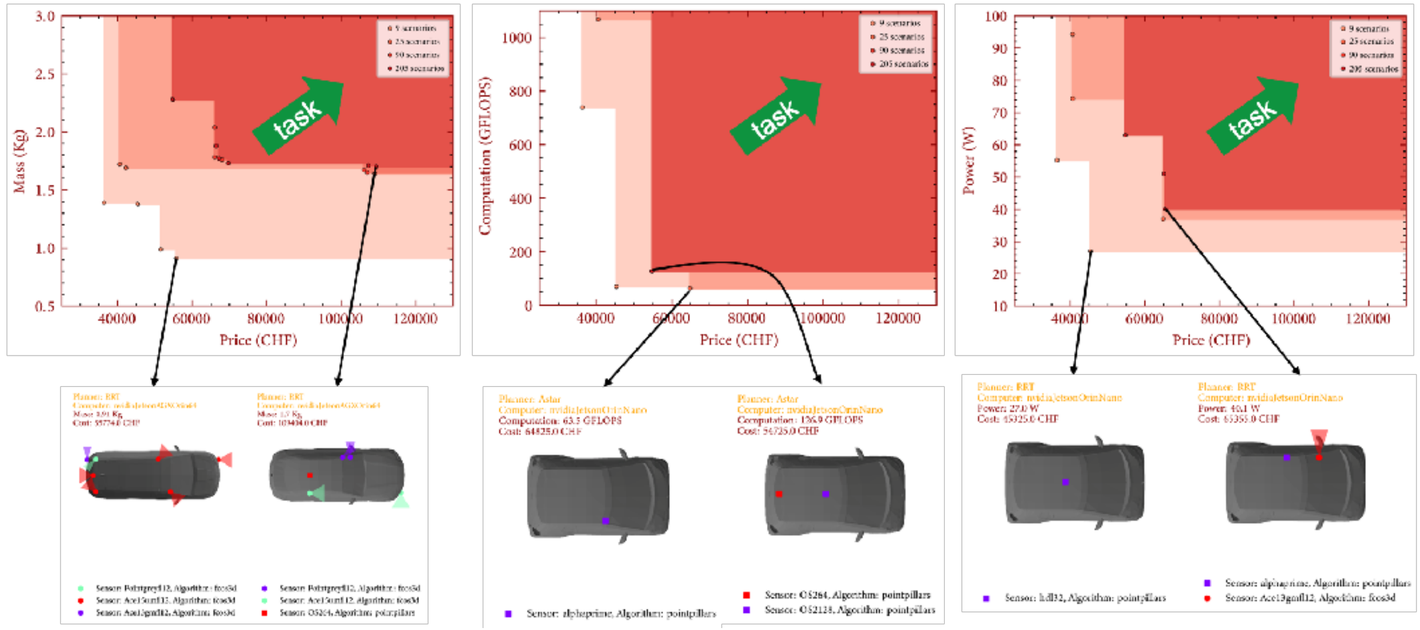


[TRO'25, CDC'22, IROS'21, ECC'21]

We can increase the granularity of our co-design models

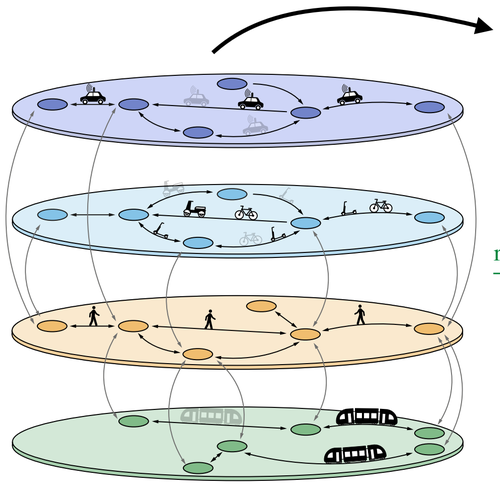


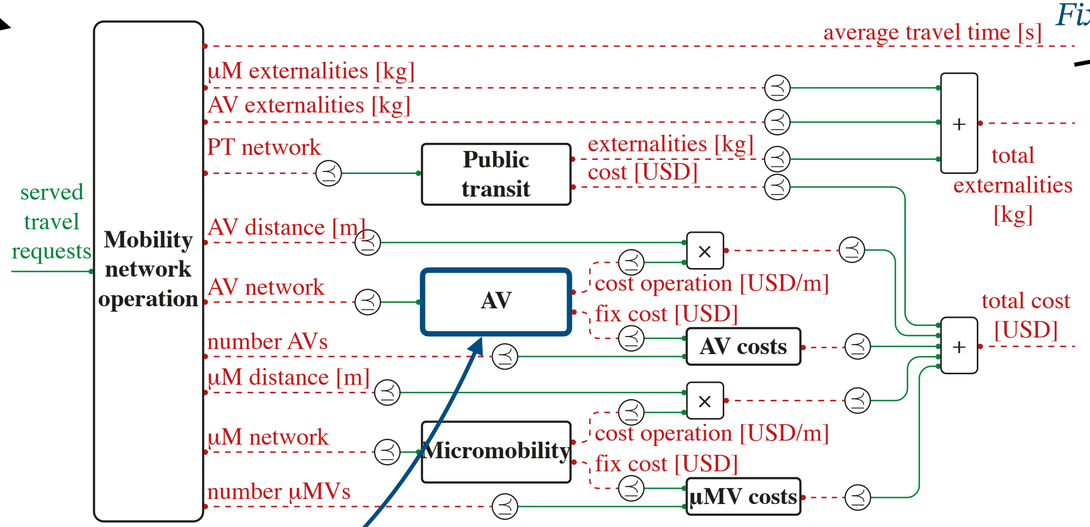




[T-RO'25]

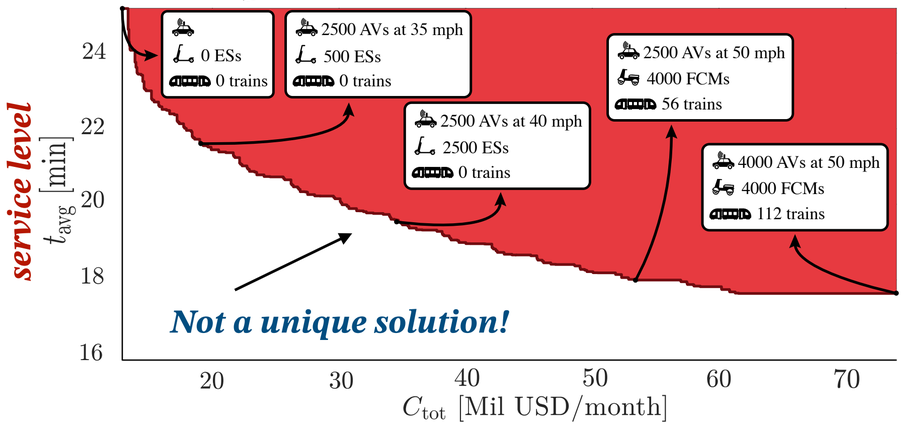
Co-design to enable user-friendly tools to assess the impact of future mobility solutions





Fix demand, minimize costs

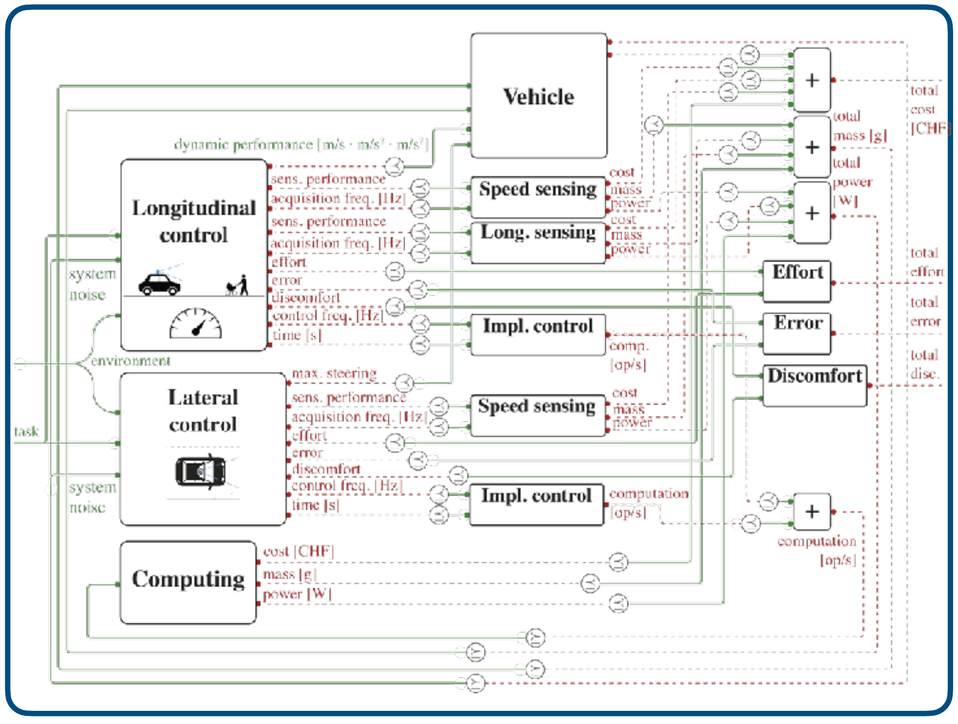
Washington D.C. in 2030



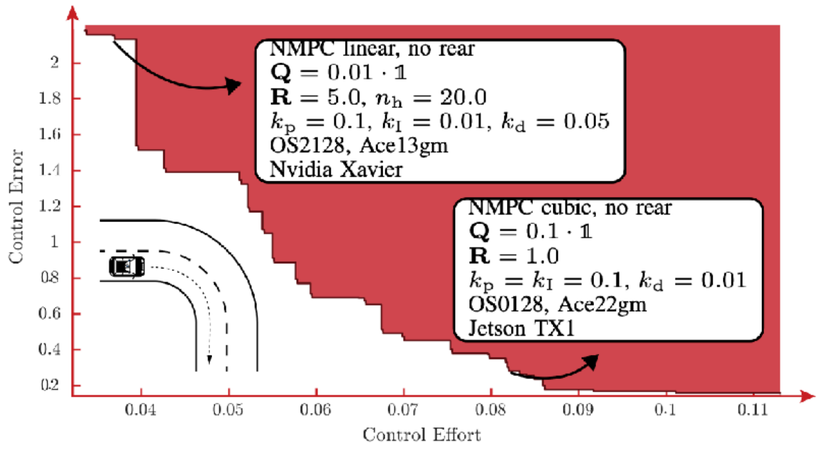
Not a unique solution!

investments

compositionality (vertical composition)



Fix environment, task



Details about software and hardware implementations, in a way that was not possible before

Sidara

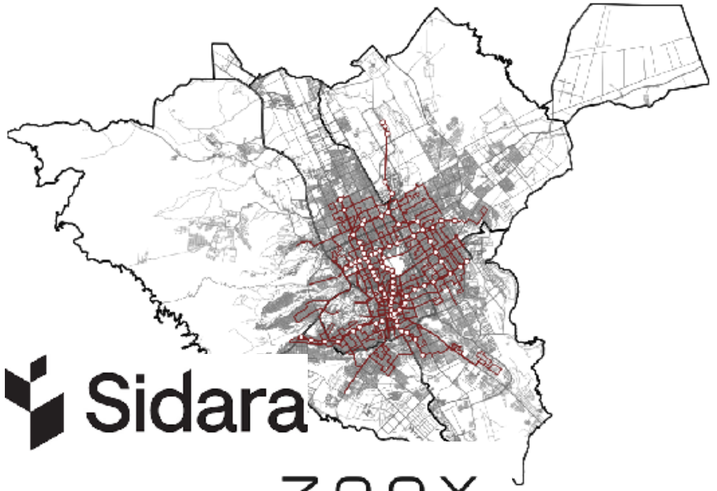
U.S. DEPARTMENT OF ENERGY

[TRO'25, TNSE'23, CDC'22, IROS'21, ECC'21, ITSC'20]

Need new tools to model and solve complex systems design optimization problems

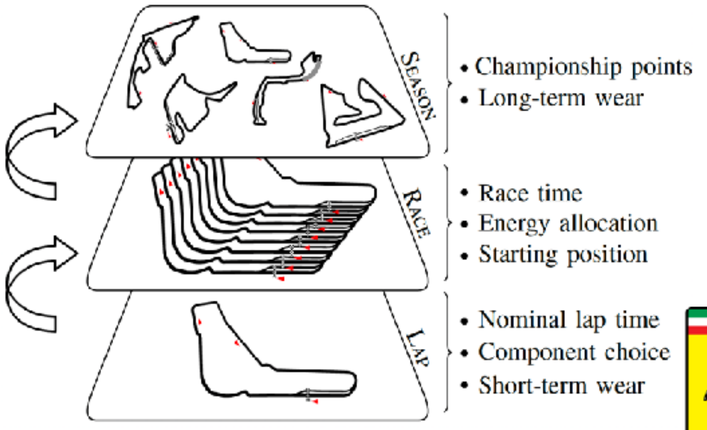
amazonrobotics

SD Guthrie

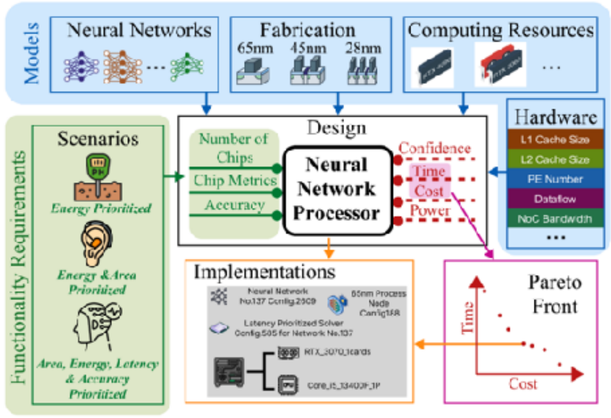


SMI Singapore Maritime Institute
Advance research. Spark innovation.

ABS

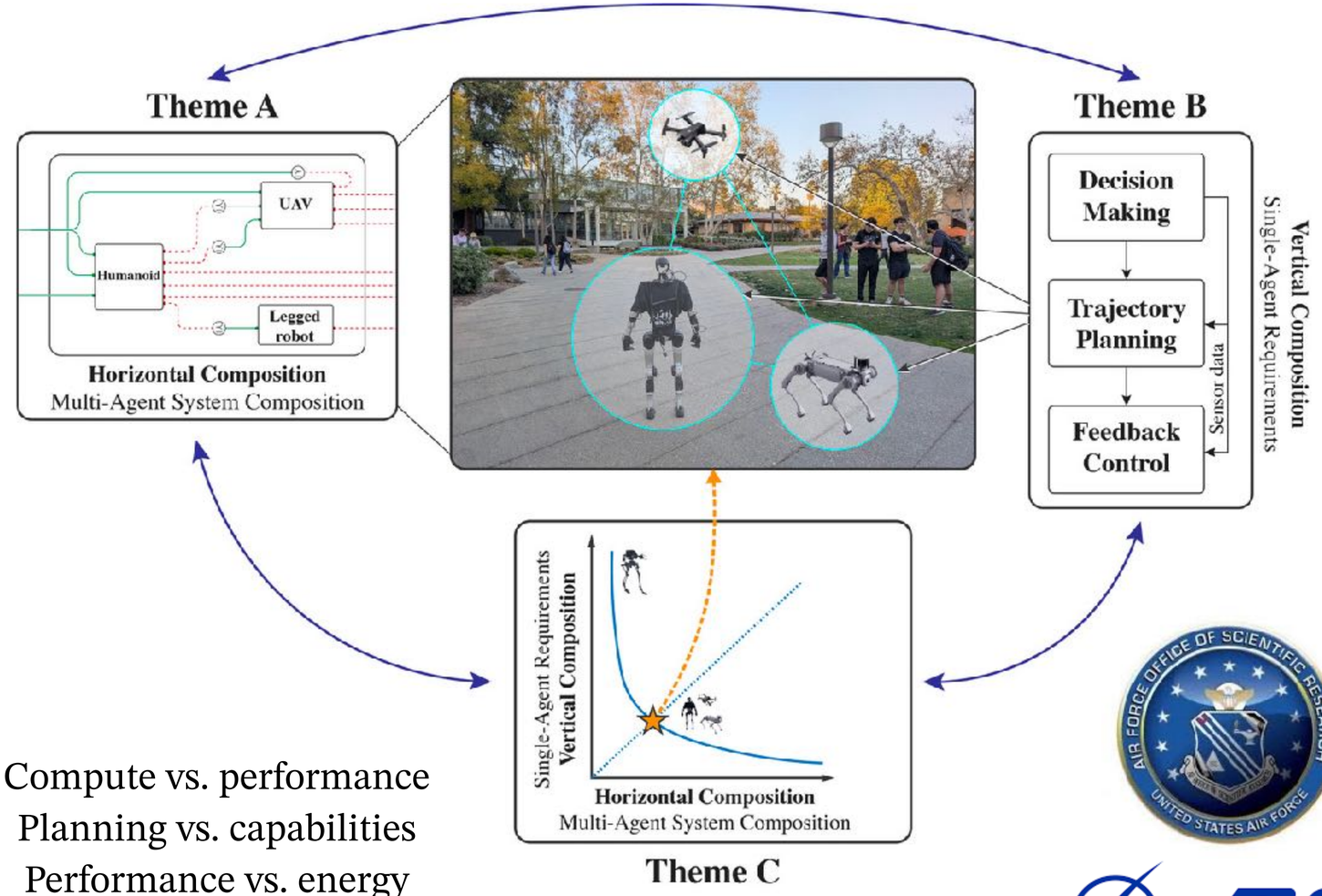


Ferrari



..and more

A new center of excellence on compositional optimization for multi-agent autonomy



Compute vs. performance
Planning vs. capabilities
Performance vs. energy

....

When???

SCHEMA
HORIZONTAL COMPOSITION
VERTICAL COMPOSITION
GLOBAL-LOCAL INTEGRATION
CODAC CoE

AIR FORCE OFFICE OF SCIENTIFIC RESEARCH
UNITED STATES AIR FORCE

BOEING

cadence

NVIDIA

Carrier

LOCKHEED MARTIN

Fleet and orchestration co-design for multi-agent autonomy

- You want to **cover a certain area** and are deciding the **fleet** and the **orchestration (planning) algorithm**
- You care about **time**, **energy**, **cost**, ...

Meshal Alharbi

Yujun Huang

Max Stralz

Fleet and orchestration co-design for multi-agent autonomy

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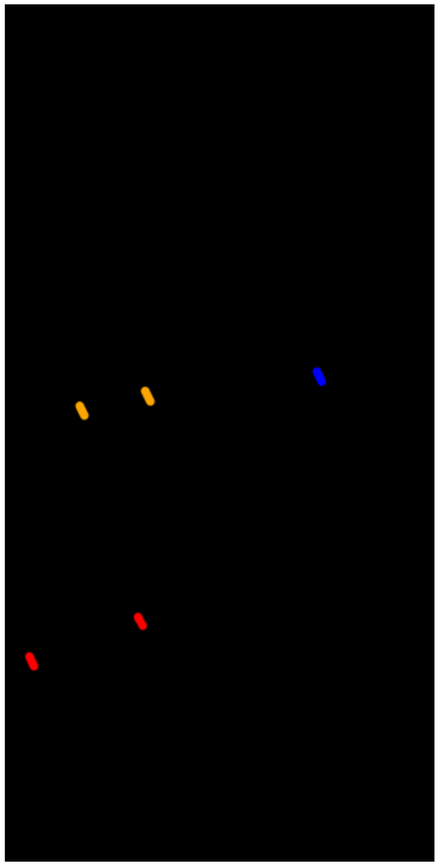
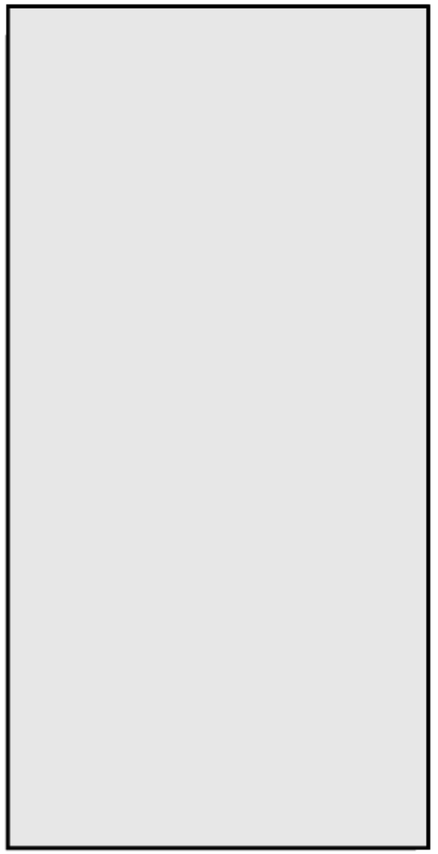
- You care about **time**, **energy**, **cost**, …

- Various planners, characterized by:
  - Computation needs
  - Scalability w.r.t. fleet sizes/areas
  - Different logics
  - Performance, parallelizability, etc.

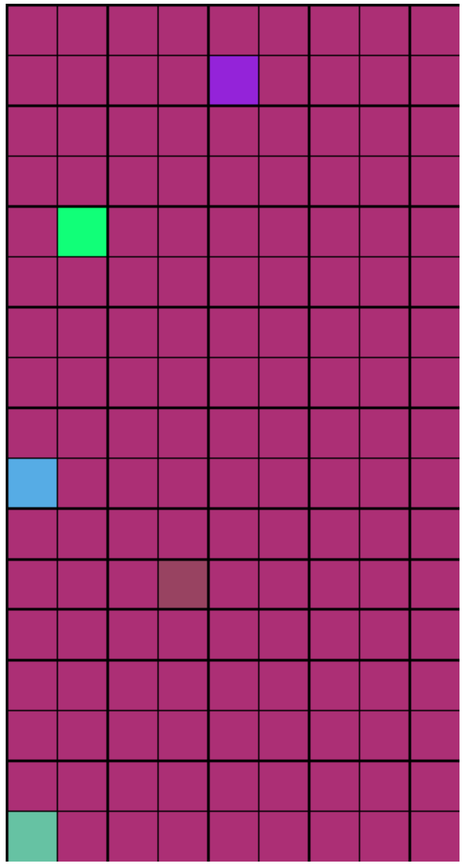
- Various parameters for a fleet:
  - Fleet **size** (how many robots)
  - Fleet **composition** (how many of each)
  - **Batteries, sensing, perception, compute, actuation**, etc.



MRTA



AGD



DARP

Meshal Alharbi

Yujun Huang

Max Stralz

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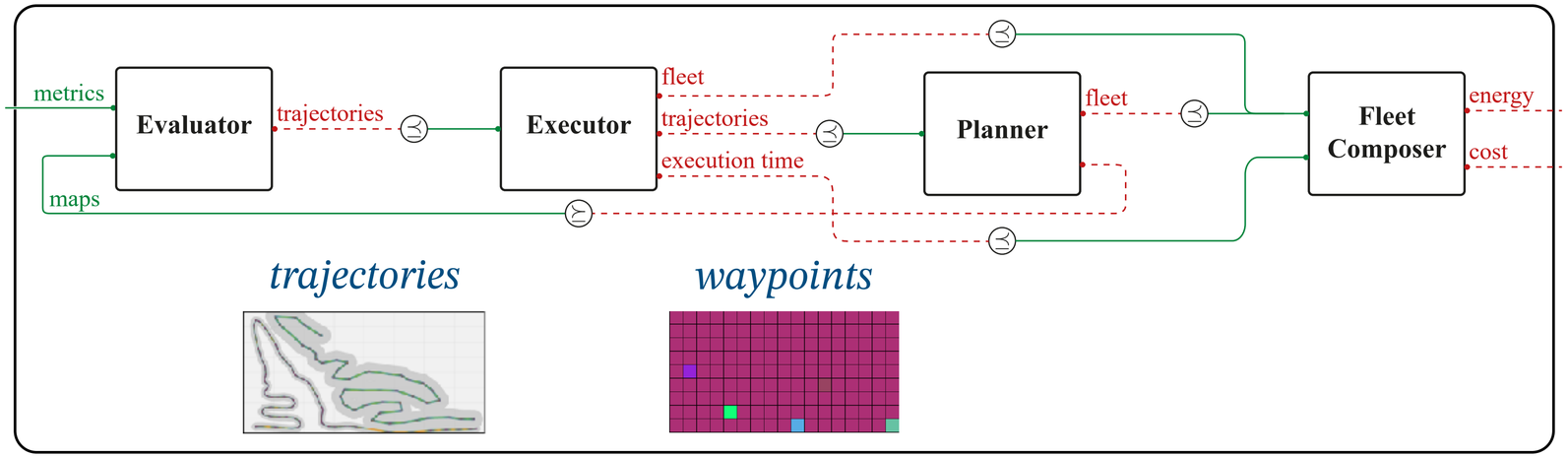


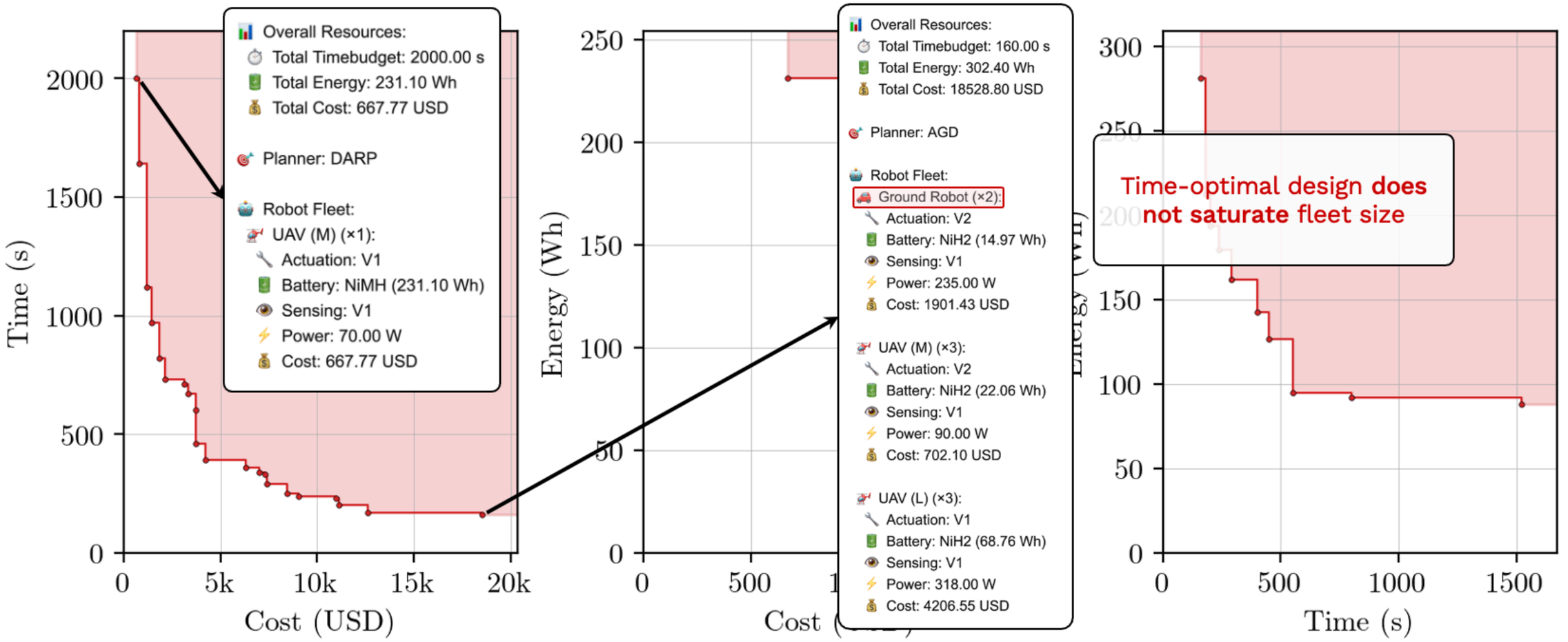
Meshal Alharbi

Yujun Huang

Max Stralz

.... of course, it's a co-design problem!

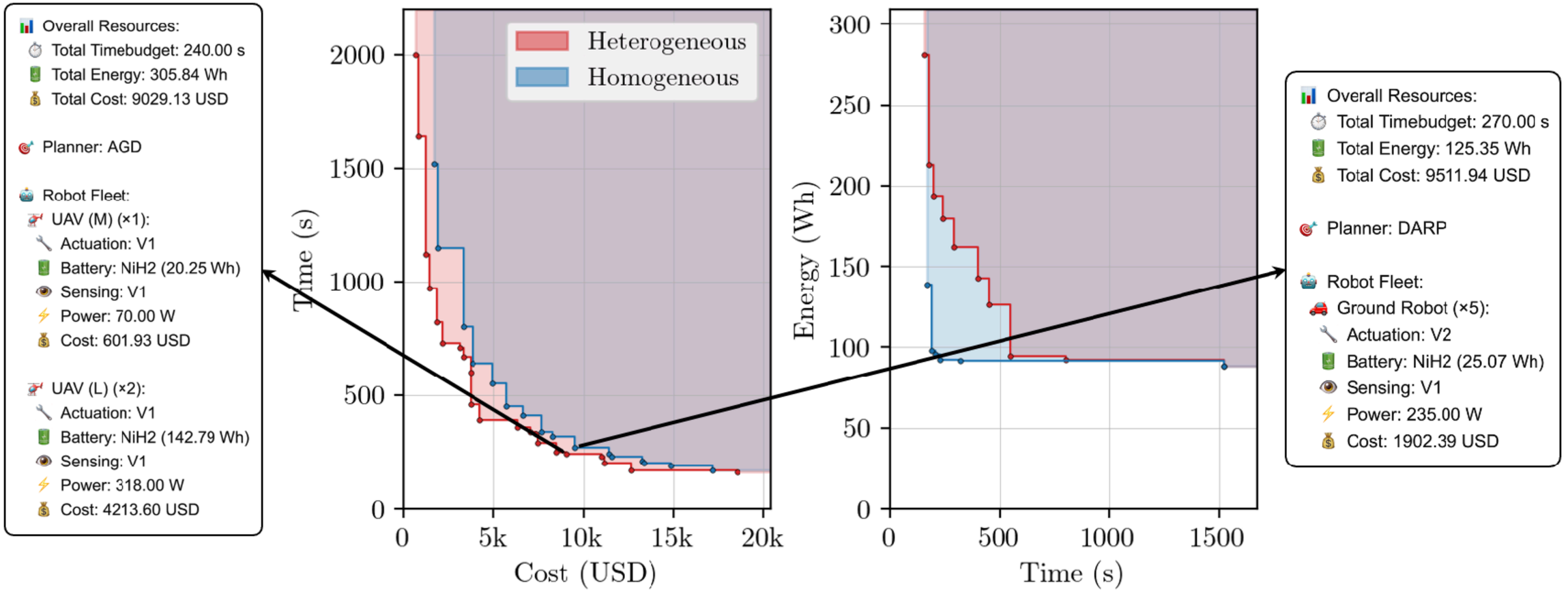
Fleet and orchestration co-design for multi-agent autonomy



… again, all the way from fleet design, to coordination

[TRO'26 (UR)]

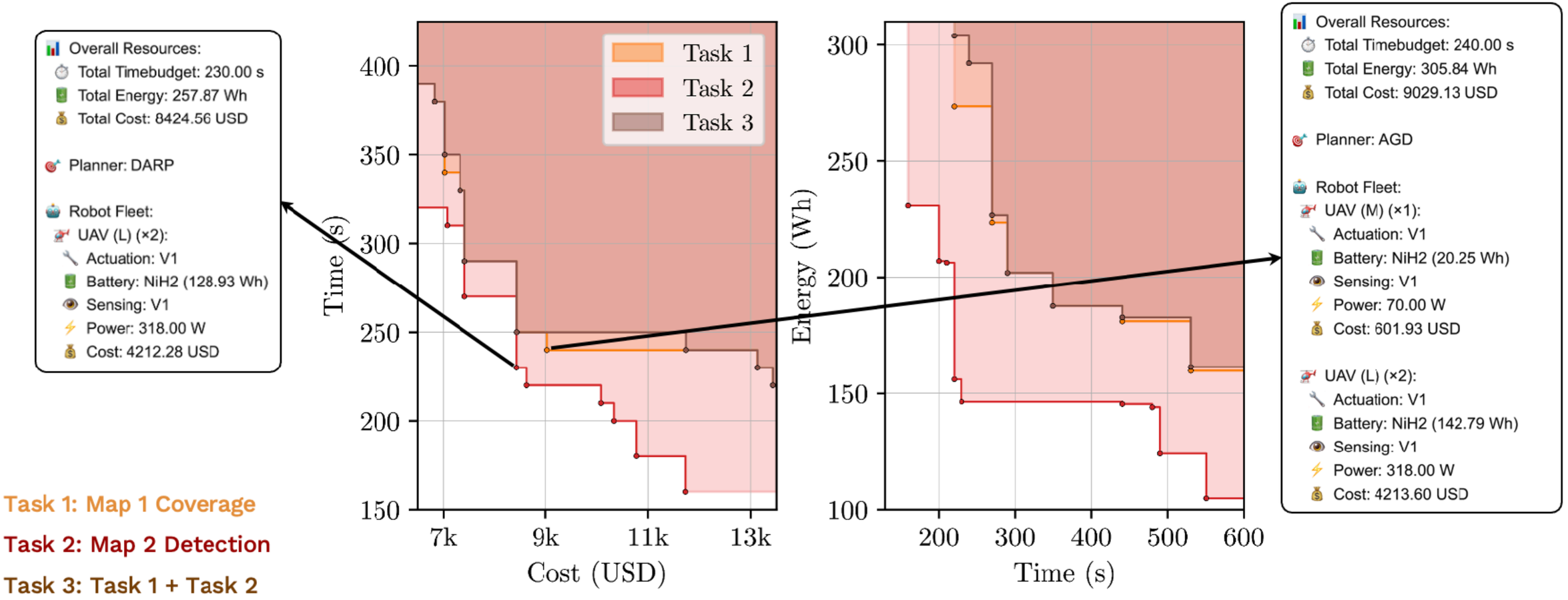
Fleet and orchestration co-design for multi-agent autonomy



Heterogeneous vs. homogeneous fleets

[TRO'26 (UR)]

Fleet and orchestration co-design for multi-agent autonomy



Task 1: Map 1 Coverage

Task 2: Map 2 Detection

Task 3: Task 1 + Task 2

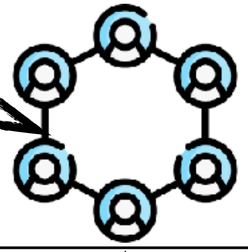
Design under task uncertainty

[TRO'26 (UR)]

Distributed Collaborative Learning and Designing of Complex Systems

Interactions between stakeholders

Distributed stakeholders across disciplines



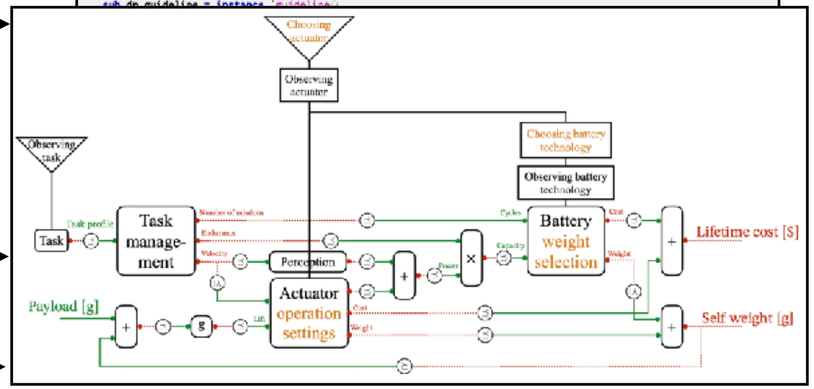
Negative (infeasibility) information

Partial knowledge and inherent uncertainty

```
ncdp {
    # Functionalities of outside "big" DP
    provides satisfied_demand ['satisfied_demand']

    # Requirements of outside "big" DP
    requires total_cost [$]
    requires dissatisfaction_level [0]
    requires total_emissions [kg/year]

    # Import DPs from the project
    sub dp_fleet = instance 'fleet_new'
    sub dp_guideline = instance 'guideline'
```



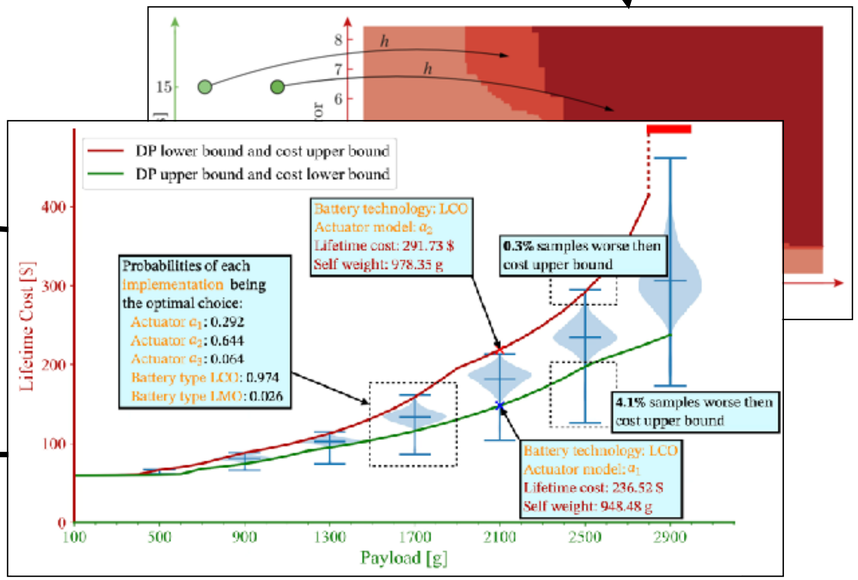
Structured Optimization

Temporal problem

Existing tools

Posterior estimation & model validation

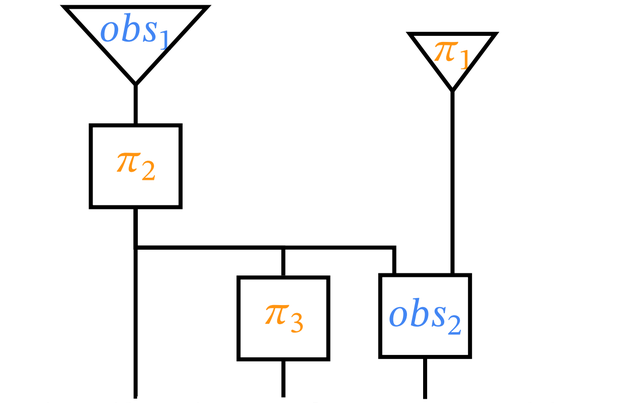
Risk aware trade-offs



Exploration and exploitation aware sampling and experiment design

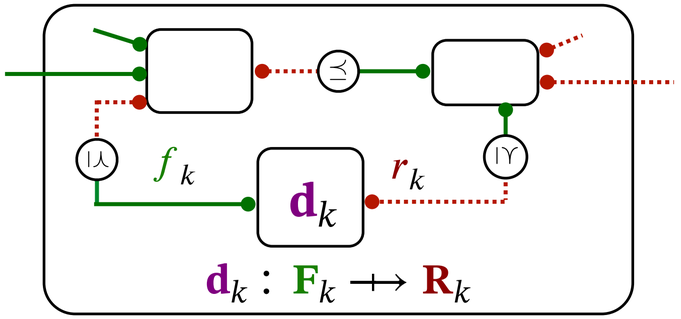
Compositional Modeling of Uncertainty and Adaptive Design Processes

- Categorical model for uncertainty × Co-design = Composable parameterized uncertain co-design
  - Compositional properties again **guaranteed**



Distributions of Design Problems

×



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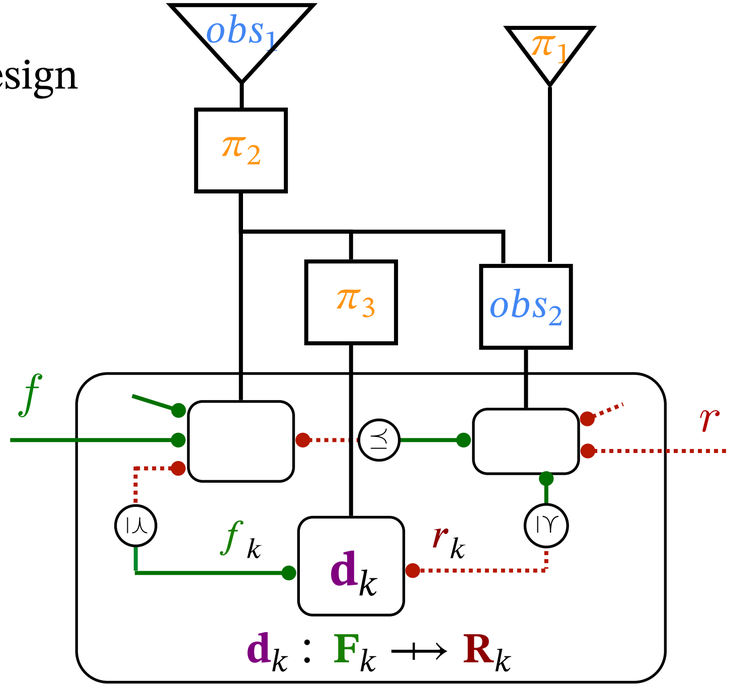
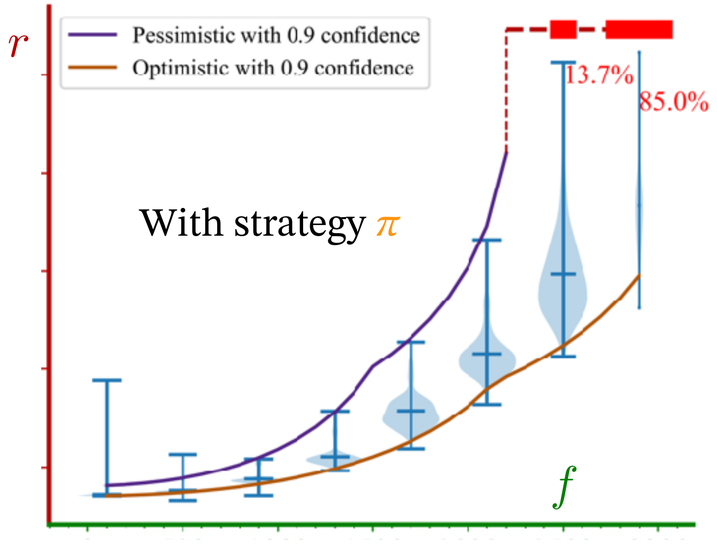


- Stochastic optimization: best decision with uncertainty
- Efficient and functorial queries
- Adaptive sampling of problems and parameters
- Learning parameters from experiments
- Exploration and exploitation trade-off
- ……



Meshal Alharbi

Yujun Huang

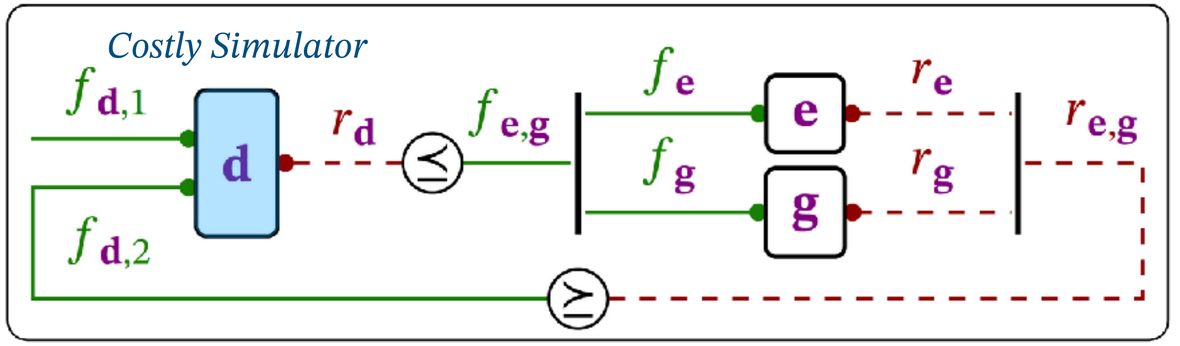
Marius Furter

[ACT'25, CDC'25, TAC'26 (UR)]

Adaptive Learning in Systems Co-Design

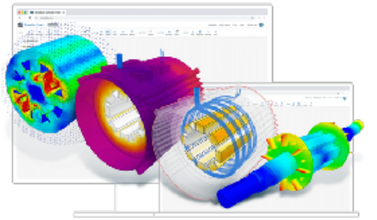
Meshal Alharbi

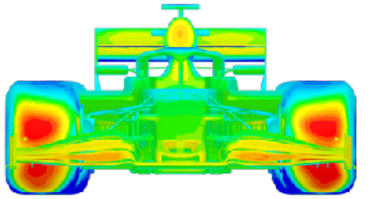
- Online learning $\times$ Co-design = Adaptive sampling to solve co-design queries with embedded **black-box models**
  - Keeping **compositionality** while treating the costly block as an unknown interface



Expensive simulations/optimizations,

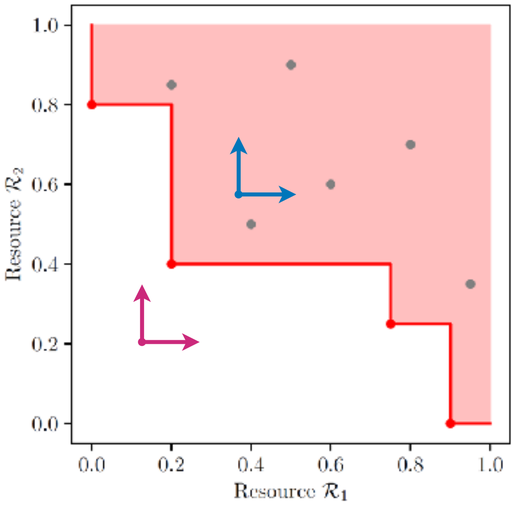
Real-world experiments





- **Optimism under uncertainty:** best decisions with current information
- Abstractions that are **agnostic** to choices for **functionality** and **resource spaces**
- **Propogating optimisim** to reason about the **system-wide utility** from the potential **local component utility**
- …

Should we sample i or j ?



History-dependent optimistic evaluation *to reason about which implementations are* ***"safe" to eliminate***

$$\mathsf{req}^{\mathrm{opt}}(i, H) \preceq_{\mathcal{R}} \mathsf{req}(i)$$

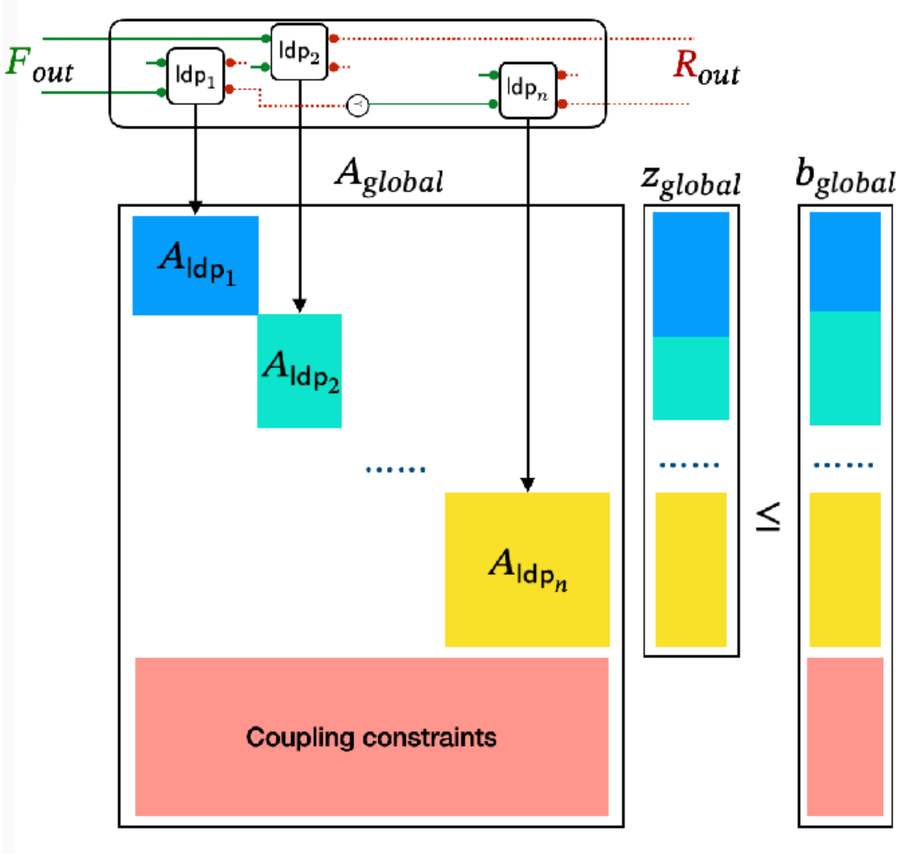
$$\mathsf{prov}(i) \preceq_{\mathcal{F}} \mathsf{prov}^{\mathrm{opt}}(i, H)$$

[TAC'26 (UR)]

Linear Co-Design Problems: Motivation and Definitions

Yubo Cai

- **From General Co-Design to Linear Co-Design**
  - Bridging Monotone Co-Design and **Polyhedral** Optimization
  - Just like LP for MP, LDP is the cornerstone for Co-Design
  - Polyhedral Approximation: Strong Expressiveness
  - More tractable algorithms



Design problem: feasible functionality/resource pairs:

$$\mathrm{dp} \subseteq F \times R$$

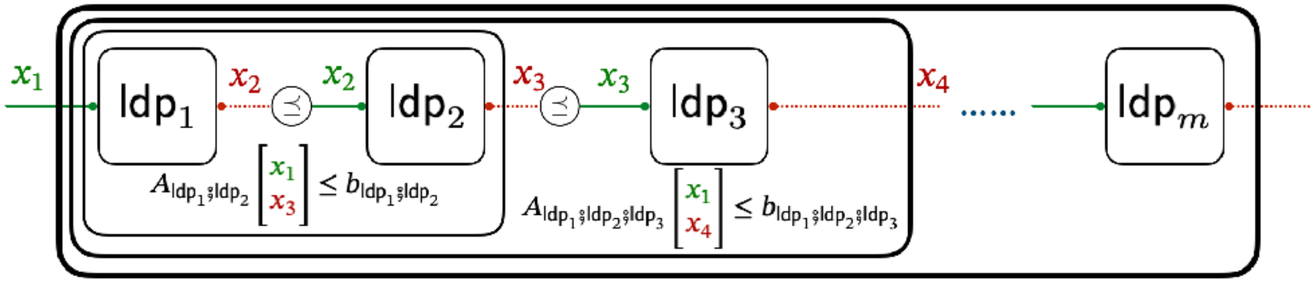
$$\min_{\preceq_R} \; x_R \quad \text{s.t.} \quad \hat{x}_F \preceq_F x_F, \; \langle x_F, x_R \rangle \in \mathrm{dp}$$

Linear constraints

Linear design problem: feasible functionality/ resource pairs as ***polyhedron***

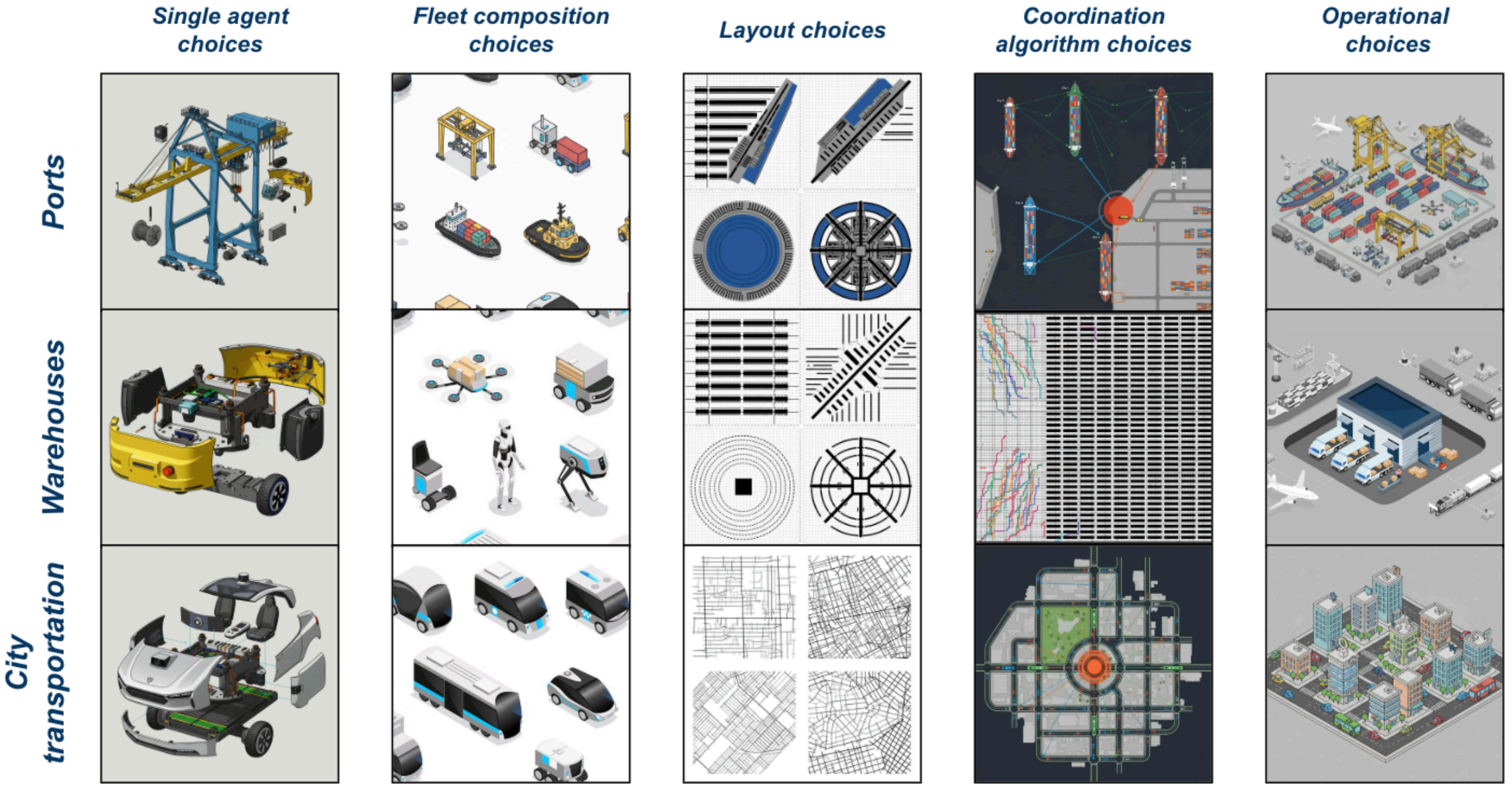
$$\mathsf{ldp} = \left\{ \langle x_{\mathbb{R}^m}, x_{\mathbb{R}^n} \rangle \mid \begin{bmatrix} A_{\mathbb{R}^m} & A_{\mathbb{R}^n} \end{bmatrix} \begin{bmatrix} x_{\mathbb{R}^m} \\ x_{\mathbb{R}^n} \end{bmatrix} \leq b \right\}$$

$$\min_{\preceq_{\mathbb{R}^n}} \; x_{\mathbb{R}^n} \quad \text{s.t.} \quad \hat{x}_{\mathbb{R}^m} \preceq_{\mathbb{R}^m} x_{\mathbb{R}^m}, \; A \begin{bmatrix} x_{\mathbb{R}^m} \\ x_{\mathbb{R}^n} \end{bmatrix} \leq b$$

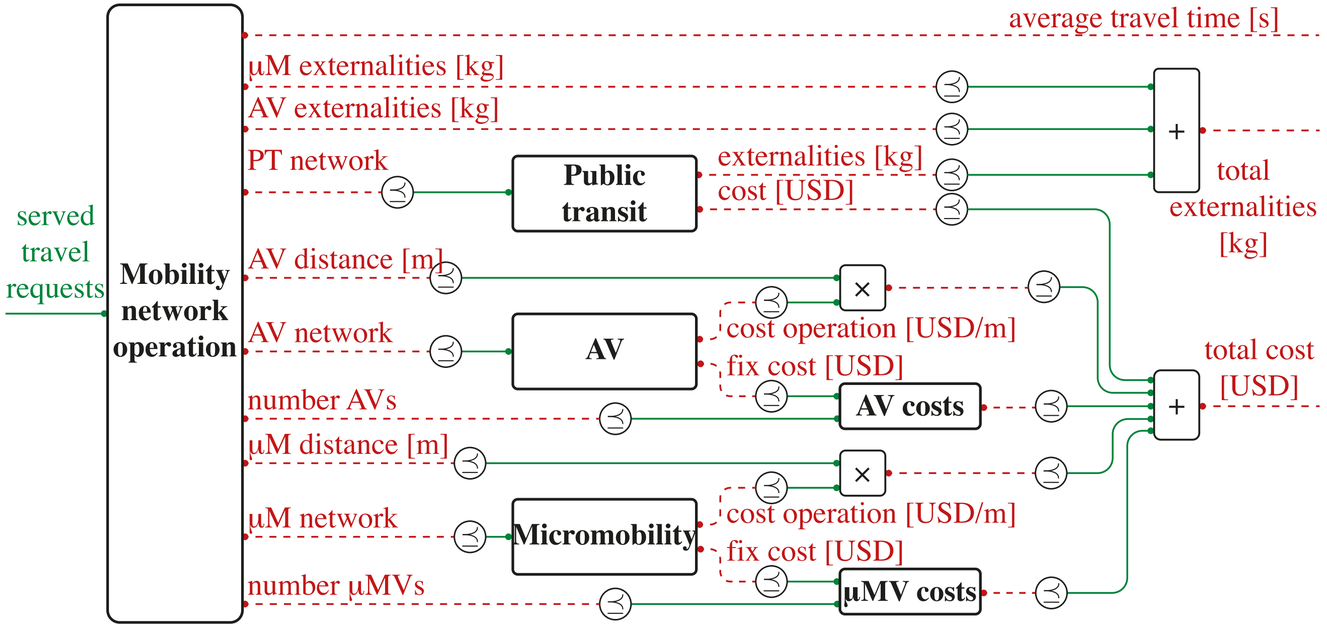


[TAC'26 (UR)]

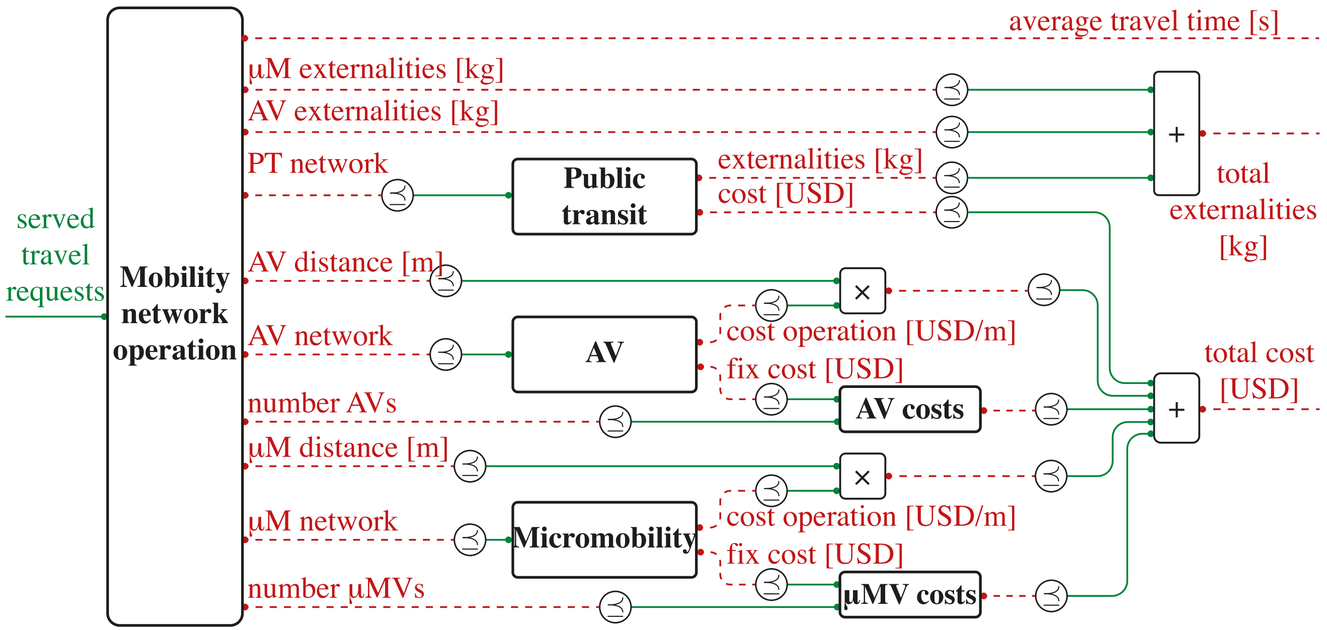
Co-Design of Networks and Operations



Explicitly accounting for strategic interactions: towards co-design games



Explicitly accounting for strategic interactions: towards co-design games



- Different **design problems** belong to different **stakeholders**
- Game theory: Multi-agent **strategic** decision making
  Allows one to **model interactions**
- The notion of **optimal designs** extends to **equilibria of designs**
- Towards a theory of **co-design games**

- Two **milestones** towards **co-design games**:

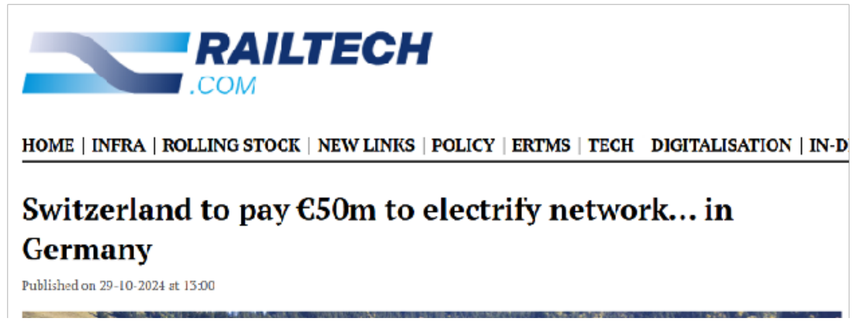
  Co-design features **rich cost structures** (posets):
  - **"Posetal Games"** (games with posetal preferences)

  [ongoing work, RA-L'23]

  Interactions are naturally **hierarchical**:
  - **Mobility games** via Stackelberg

  [ITSC'21 (***Best Paper Award***), ITSC'23, ACC'25, TCNS'25]

Co-designing Transportation Networks: Competition and Cooperation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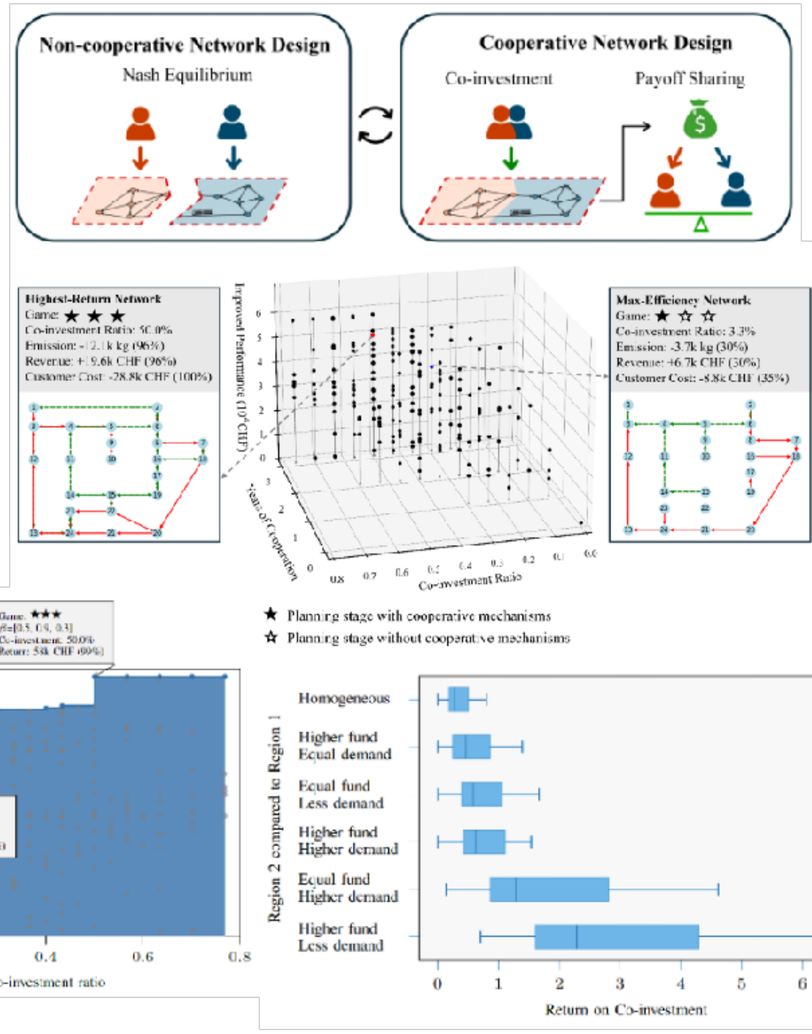
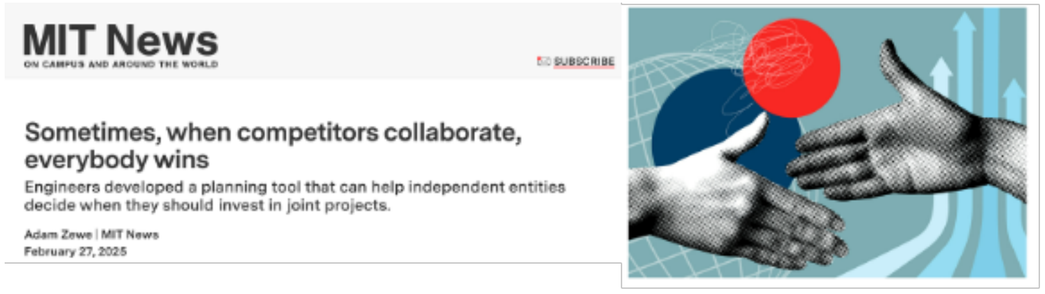
Switzerland to pay €50m to electrify network... in Germany

Published on 29-10-2024 at 15:00



- **Co-investment with Payoff-sharing Mechanism**
  - Integrating non-cooperative and cooperative game theory
  - Fostering mutually beneficial cooperation in competitive environments
  - Decision-making tools for cooperative strategies among self-interested operators

- **Experiments on real-world transportation networks**
  - Early investment yields **significant long-term benefits**
  - Higher co-investment may not lead to high returns
  - **Diversity** enables larger opportunities


MIT News

ON CAMPUS AND AROUND THE WORLD

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Sometimes, when competitors collaborate, everybody wins

Engineers developed a planning tool that can help independent entities decide when they should invest in joint projects.

Adam Zewe | MIT News

February 27, 2025

Agenda

- **Motivation**
  - *New challenges of engineering design*
  - *Motivation from autonomy and mobility*
  - *Desiderata for co-design*
- **Monotone Co-Design**
  - *Modeling design problems*
  - *Examples across domains*
  - *Design queries and optimization*
  - *From autonomy to mobility systems*

- **Outlook on ongoing and future research**

Website containing all papers and more pointers:
https://zardini.mit.edu

Modeling & Algorithmic Foundations

Societal Applications

User-friendly Tools

My lab is building the next generation tools for systems design optimization

Modeling and Algorithmic Foundations

Leveraging **optimization**, **control theory, game theory, domain theory,** and **applied category theory**:

- Extend and improve current **modeling** & **solution algorithms** for **multi-objective** design optimization
- Promote **interdisciplinarity** by bridging the gap between **standard optimization** and **co-design**
- Explicitly account for **strategic interactions** of stakeholders, developing a theory of **co-design games**

Societal Applications

Mobility, networks, infrastructure
Strategic interactions at all levels

Mission-driven autonomy

Aerospace, automotive, production chains, energy and data networks

User-friendly Tools

Collaborative, intellectually tractable

Authorities & Industry

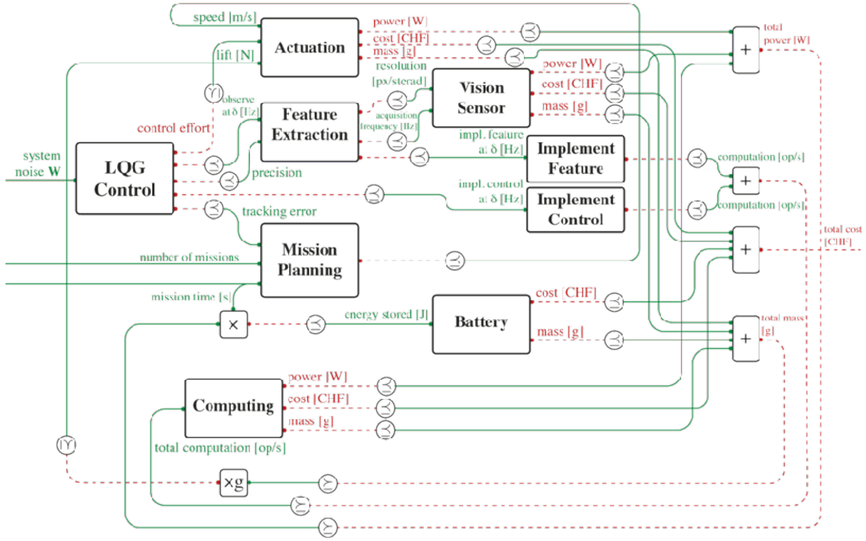
Literature, workshops, classes

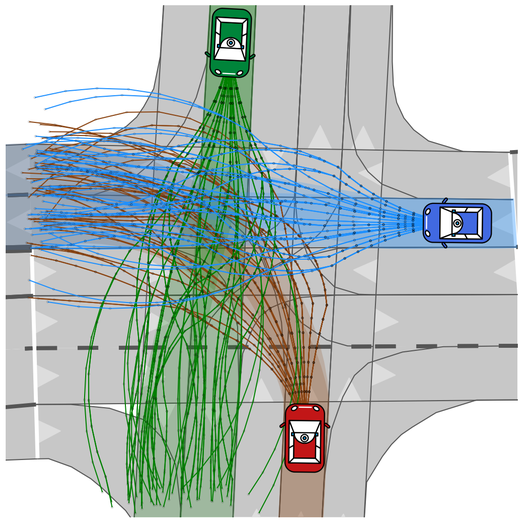
Take-aways

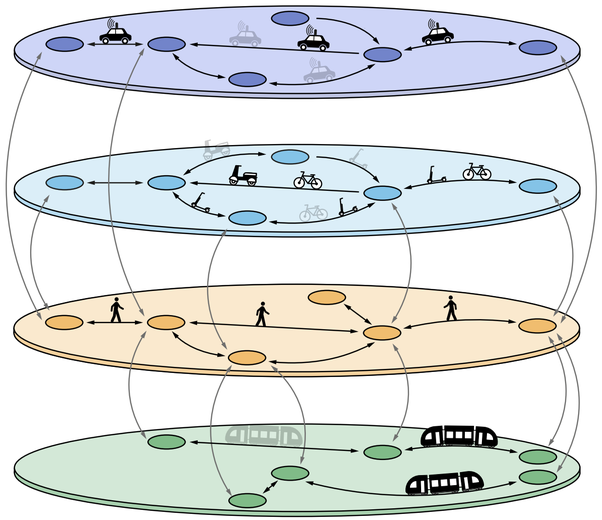
- A new approach to **co-design** designed to work **across fields** and **scales**.
- It is:
  - **Compositional** *horizontally* and *hierarchically*.
  - Supports both **data-driven** and **model-based** components.
  - **Computationally tractable**.
  - **Intellectually tractable.**
- Currently: extend **modeling** and **algorithmic** capabilities
- Currently: exploring **strategic aspects**

- Main applications: **multi-agent autonomy, society-critical networks**

Access the book at:
https://bit.ly/3qQNrdR







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Questions?

I'm hiring/welcoming visitors

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